Primer on Phylogenetic Comparative Methods for the Biological Sciences

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Preface

What are comparative methods?

The comparative approach is one of the oldest and most frequently used methods for answering questions in biology. The rationale of this approach is to test biological hypotheses by comparing organisms that differ in their life history, ecology or evolutionary history. Although originally developed in the 18th century as a qualitative approach for describing biological diversity, the modern comparative method is highly quantitative, drawing together methods from phylogenetics, advanced statistics, and computational biology. The comparative method is used in all areas of biology, including ecology, evolutionary biology, behavioural ecology, conservation biology, anthropology, primatology, epidemiology, and physiology.

The modern comparative method recognises that robust tests of biological hypotheses require that we account for the evolutionary relationships among organisms. Closely related species share adaptations and ecological preferences because they inherited them from their common ancestor, not because of independent evolution (phylogenetic non-independence). The outcome of this is that statistical methods that assume that species data are independent may be compromised. For this reason, the past 40 years has seen a huge effort devoted to developing statistical methods that are robust to this problem.

Although much of the development of the comparative approach has revolved around statistical tests, more recently the scope of new methods has expanded to consider the mode and nature of evolution. Recognising that the level and nature of phylogenetic non-independence is a function of trait evolution, new methods have been developed that permit the measurement of rates of trait evolution, including time-varying rates, clade dependent evolution and relationships between traits and diversification.

Why did we write this Primer?

For a novice, the field of comparative methods is a potential minefield. There is >40 years of literature, including a great deal of technically intense methods development. Most potential users of comparative methods are familiar with basic statistics and some advanced methods (e.g. generalised linear models; GLMs), however the comparative method can be daunting and difficult to understand, because the methods are statistically advanced, and require some understanding of phylogenetic concepts that many non-specialists will be unfamiliar with. This *Primer* is designed to help new users understand and use comparative methods.

Who is this Primer for?

This *Primer* is pitched at an audience ranging from advanced undergraduates to early postdoctoral researchers. No previous knowledge of comparative methods is assumed, however we assume a general understanding of evolutionary biology, particularly that we can represent evolutionary relationships as phylogenetic trees. We assume some knowledge of basic statistics such as regressions, and refer to other commonly used statistics, e.g. GLMs, throughout. However, these references are designed to aid readers familiar with these techniques; if readers are unfamiliar with them it will not substantially reduce their understanding. For those wanting a more thorough coverage of particular methods or models we have included references to further reading.

How is this *Primer* organised?

[TO BE COMPLETED] The *Primer* is divided into eight chapters as follows:

- Chapter 1: Introduction to the comparative method.
- Chapter 2: Foundations 1: .
- Chapter 3: Foundations 2:
- Chapter 4: Dealing with phylogenetic non-independence: a practical introduction.
- Chapter 5: Testing and evaluating comparative models: is my model good enough?
- Chapter 6: More advanced topics, where next?
- Chapter 7: Tempo and mode of evolution from comparative data.
- Chapter 8: Looking forward, more advanced comparative methods and where next.

Online companion material

[TO BE COMPLETED] Each chapter of the *Primer* has associated online companion material. This consists of 1. An online book with practical examples of how to perform each analysis in R. 2. Multiple choice questions for each chapter so you can test your understanding. 3. Short videos.

Boxes

Throughout the *Primer* are boxes of text highlighting particularly important issues:

Note boxes. These boxes highlight important details that we don't want you to miss!

Extra details boxes. These boxes contain detailed explanations of things for readers who like to fully understand the complexities of what they are doing, for example fully worked through examples, or technical details that we have not explained in detail in the text.

Caveats boxes. These boxes highlight important points that need to be considered when working through your own analyses. They reveal areas where it is important to be careful and think about what you are doing and **why**. The image is a Jurassic Park era velociraptor to remind you of the "Jurassic Park caveat", i.e. that just because you *can* perform an analysis in R doesn't mean that you *should* (thanks to Dr Ian Malcolm and Dr Michael Crichton)! Always consider the question at hand, your study group, and the quality of the data you are using before embarking on a new comparative analysis.

Acknowledgements

People who read drafts etc etc.

Chapter 1

Introduction to the comparative method

** RPF is writing this at present **

Here we will lay the conceptual framework for the comparative method. The idea behind the comparative method will be introduced, and examples given of the types of analysis that can be undertaken. The problem of phylogenetic non-independence will be introduced and a range of methodological approaches described in brief.

- Comparison as a method to test hypotheses.
- Historical context of the comparative method (i.e. pre Darwin to today).
- What questions can we answer?
- The problem of phylogenetic non-independence: understanding the conceptual problem caused by the way evolution occurs, as well as the statistical consequences.
- Key examples (e.g. early examples; examples where phylogeny makes a difference).
- A taxonomy of methods: Table of methods with links to further reading.

Chapter 2

Foundations 1: introduction to phylogenies and comparative data

[CHAPTER OPENING STYLE STILL TBC AND WILL BE CONSISTENT ACROSS CHAPTERS. CURRENT CONTENT IS PLACEHOLDER ONLY]

Before we use comparative methods, we need to understand some basic concepts about phylogenies, comparative data, and the models underlying them. In this chapter we will cover:

- Phylogenies.
  - Basic introduction.
  - How are phylogenies inferred?
  - Reading and using phylogenies.
- Comparative data
  - basics.

2.1 A basic introduction to phylogenies

At its simplest, a *phylogeny* or *phylogenetic tree* is a branching diagram that shows how species are related. The ancestor of all of the species is found at the *root* of the tree, with the species at the *tips*. The rest of the tree is made up of *nodes* and *branches*. Nodes show us where *speciation* events have occurred, i.e. where one species has evolved into two (or more) new species. The branches show the relationships among nodes and tips (Figure 2.1).

Most phylogenies are *rooted*, meaning that the ancestral node or *root* is identified.

Phylogenies are rooted using an *outgroup*. The outgroup can be one or more species, and represents a species or group known to lie outside the group of interest, the *ingroup*. The root lies on the node that joins the outgroup to the ingroup. In an *unrooted* tree, no root is defined, meaning while we can look at the relationships among species, we cannot identify ancestral characteristics or how species change through time. This limits their usefulness, especially for comparative methods.

Some phylogenies only show the *topology*, i.e. the shape of the relationships among the species. Most phylogenies, however, will have some information about the timing of branching events. In these phylogenies, *branch lengths* can represent absolute or relative time elapsed, or amounts of character change. Most of the phylogenies we use will be *dated phylogenies* (or *cladograms*), where the branch lengths represent divergence times.

Note that throughout the *Primer* for simplicity we generally refer to "species" at the tips of phylogenies. In some cases, the tips will actually be *genera* or *populations*, or other operational taxonomic units (*OTUs*). Before beginning a comparative study, remember to check what the tips are in the phylogeny you are using.

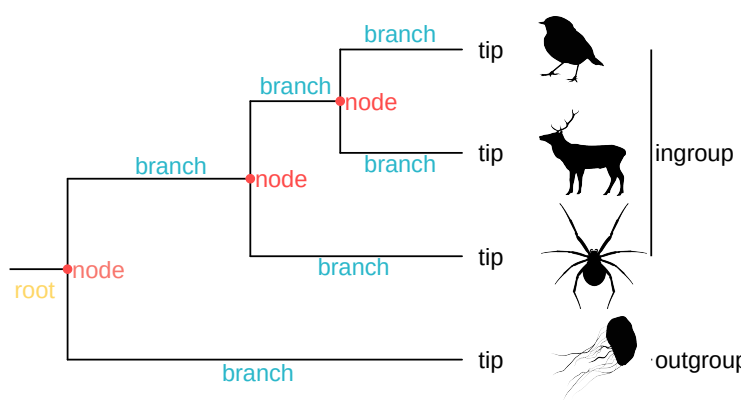

Figure 2.1: A simple phylogeny demonstrating the key features of phylogenetic trees.

To use phylogenies we need to understand a few more terms. These are also described in the Glossary (Chapter 9) if you forget these at any point.

Ultrametric. Most of the phylogenies we will use in this book are *ultrametric* where all the branches end at the same place - the present-day (Figure 2.2A). If you are using phylogenies containing extinct species then the phylogeny will usually be *non-ultrametric* (Figure 2.2B). A key thing to remember when using phylogenies, especially ultrametric phylogenies of living species only, is that extinct species are missing. Exactly how large a problem this might be for

comparative methods is unclear, because we can never know how many extinct species we are missing due to the patchiness of the fossil record. But it something we need to consider.

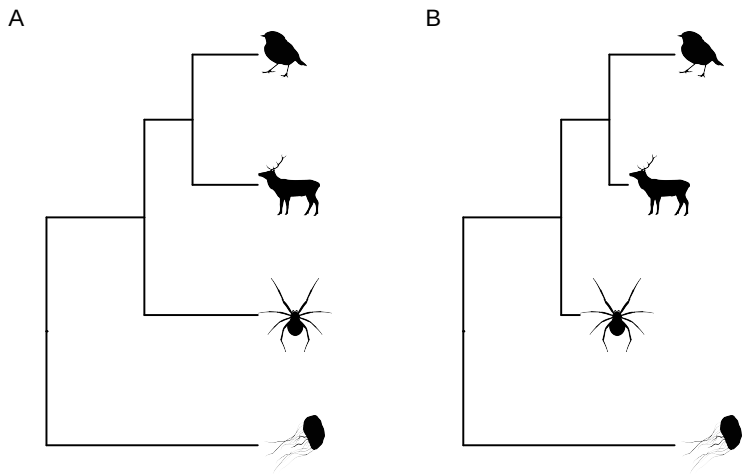

Figure 2.2: Ultrametric (A) and non-ultrametric (B) phylogenies. In (B) the deer and spider have gone extinct before the present day.

Polytomies. The phylogenies we will use in this book will be *fully resolved*, i.e. each node will lead to two branches. This might not always be the case, however. When more than two branches originate from a node we call this a *polytomy* (Figure 2.3B). Polytomies can be hard or soft. *Hard polytomies* represent the true evolutionary history of a group, often occurring where there has been rapid evolution. *Soft polytomies*, on the other hand, represent problems in phylogenetic inference (see below) where the methods cannot resolve the relationships among the species at that point.

Monophyly and clades. A group of species that all share a common ancestor are referred to as a *monophyletic group* or a *clade* (Figure 2.4). Often, common taxonomic groupings e.g. mammals, are also monophyletic groups. Sometimes, taxonomic groups are nested within other taxonomic groups in a phylogeny, for example birds are nested within reptiles so that although birds are a monophyletic group, reptiles are a *paraphyletic group* (Figure 2.5). We can also have *polyphyletic groups* where the group members appear in multiple lineages across a phylogeny, for example the group formed by birds and mammals (Figure 2.5).

2.1.1 Reading and interpreting phylogenies

We can read/interpret the phylogeny in Figure 2.6 as follows. **A** and **B** are each others' closest relatives. They share a *common ancestor* at node **X**. We can think of the branch leading to node X as the shared evolutionary history of

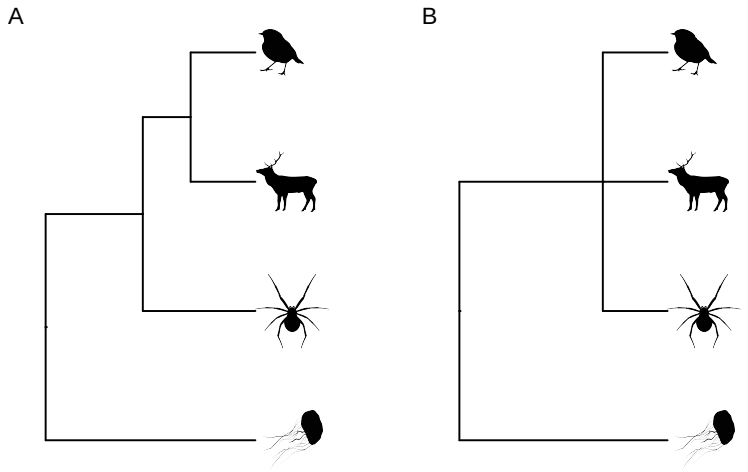

Figure 2.3: A fully resolved phylogeny (A) versus one with a polytomy (B). This polytomy suggests that the relationships among the deer, the robin and the spider are unresolved.

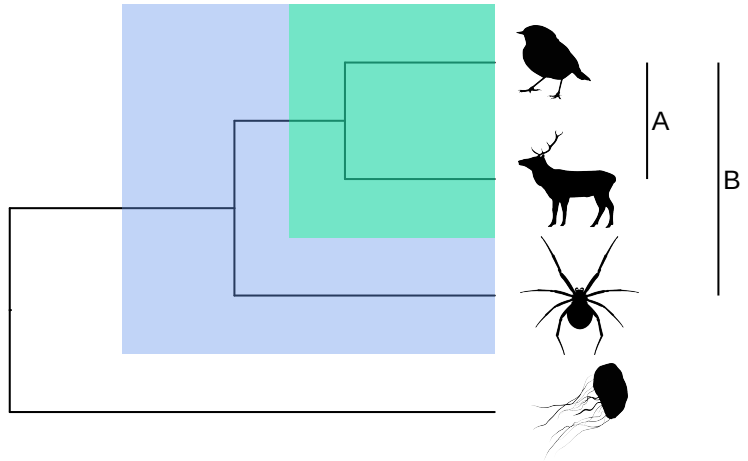

Figure 2.4: Phylogeny highlighting the monophyletic groups or clades. A is a clade made up of the deer and the robin. B is the more inclusive clade including the deer, the robin and the spider.

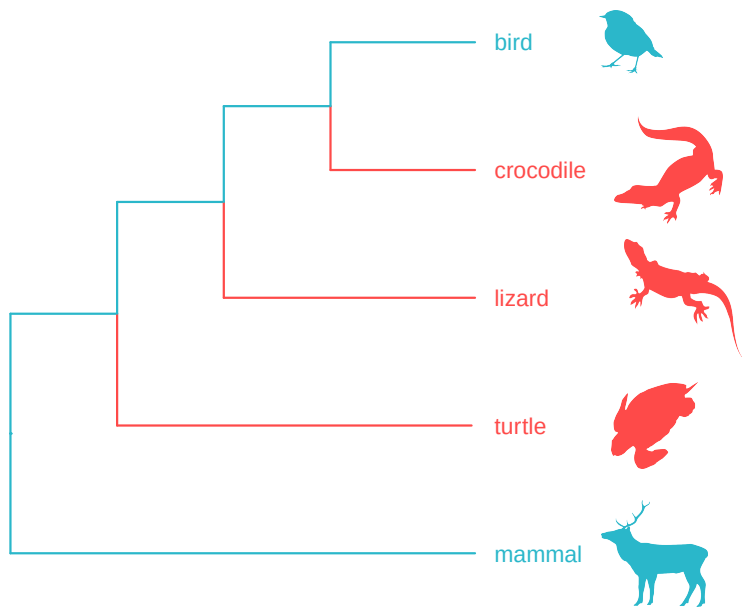

Figure 2.5: Phylogeny showing the relationships among birds, mammals and reptiles. Reptiles forms a paraphyletic group because birds is nested within it. The group formed by birds and mammals is a polyphyletic group.

A and B, while the branches leading from node **X** to **A** and **B** represents their independent evolution after speciation. **A** and **B** form a monophyletic group, or clade. They are also *sister species*.

Both **A** and **B** are equally closely related to **C**, as they all share a common ancestor at node **Y**. Note that **B** is **not** more closely related to **C** just because it is next to it at the tips. We could also represent the relationship as shown in the second panel with the branch leading to **B** and **A** rotated.

Reading phylogenies is not always intuitive. Often this is due to how we represent them graphically. For example, each of the phylogenies in Figure 2.7 shows the same relationships among species, but in different ways. By convention we usually present phylogenies with the **root at the left or the base of the phylogeny** (like A, C, E, or G in Figure 2.7).

We can also use circular or fan shaped phylogenies (Figure 2.8) when we have large phylogenies that are hard to fit onto a page.

If you find reading and understanding phylogenies difficult, we highly recommend trying the *Tree Thinking challenge* at `https://science.sciencemag.org/content/sci/suppl/2005/11/07/310.5750.979.DC1/Baum.SOM.pdf` [@Baum979], which should help you get

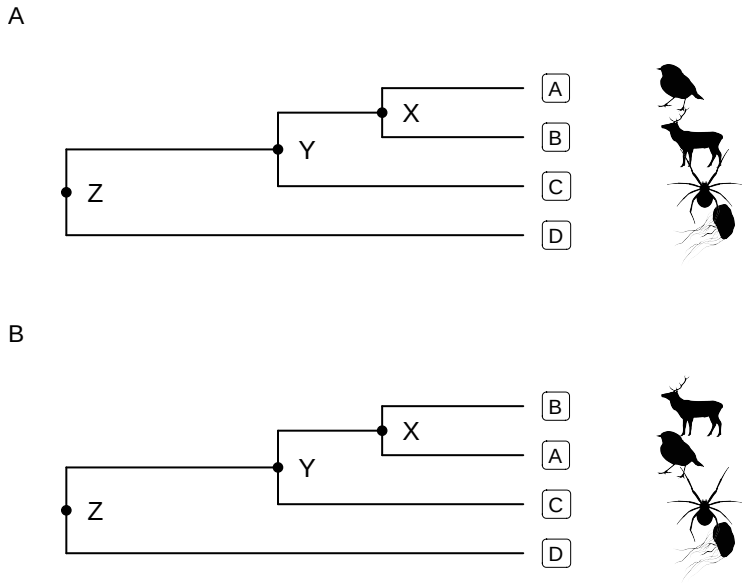

Figure 2.6: How to read a phylogenetic tree

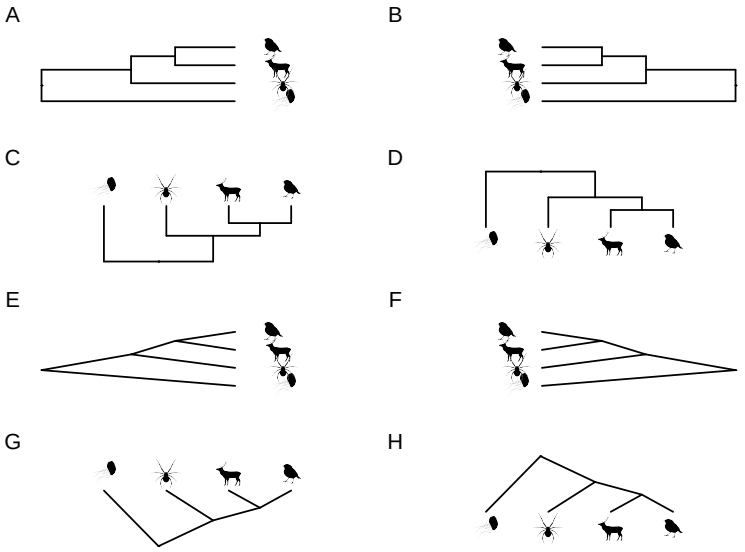

Figure 2.7: All of these phylogenies show the same species relationships, they are just presented in different ways

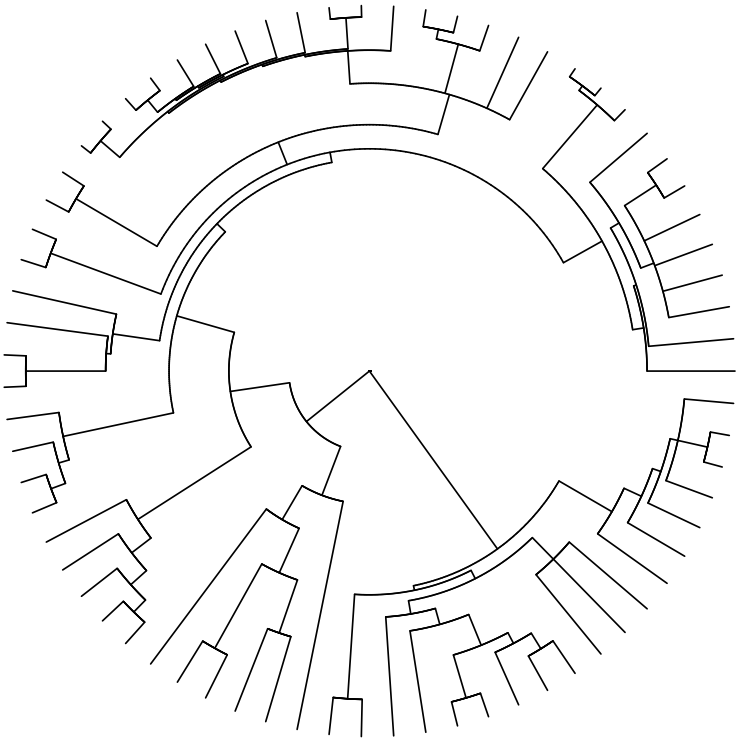

Figure 2.8: An example of a fan or circular phylogeny. Tip labels have been omitted to make it easier to see the shape of the tree.

the hang of it.

2.1.2 How do we infer phylogenies?

We could write a whole book just about inferring phylogenies. In fact, there already are several e.g. (Felsenstein, 2004), and there will be another Primer focusing on this subject if it's something you need/want more details on! Here we will just cover the basics, focusing on the information you'll need to be able to understand the rest of the book.

Note that we **infer** phylogenies. We do not build or create them. This is because whatever clever methods we use, we can never truly know the evolutionary history of most groups until someone invents a time machine!

Most phylogenies are built using genetic or morphological data. Genetic data

is usually a gene sequence, with the genes used depending on the question. For distinguishing relationships among close relatives, fast-evolving genes such as those in mitochondria or chloroplasts might be used, whereas protein-coding genes or whole genomes might be more appropriate for looking at relationships among ancient lineages. Collecting genetic data these days might be as simple as downloading it from GenBank, however, the sequences still need to be aligned which can take more time.

Morphological data is generally in the form of *cladistic matrices* where each character or trait of a species is coded as an integer. For example, a tail might be coded as 0 (absent) or 1 (present). Selecting characters for morphological phylogenies can be tricky, as they need to encompass the *synapomorphies* or unique characters that define clades in the tree, but the characters should also be independent of one another. Careful consideration of how characters should be coded is also needed, and this can vary from study to study.

Once the data have been collected they can be plugged into one of a number of software packages that will apply one or more phylogenetic inference methods to output the "best" tree or trees. Here we will just mention the three most used inference methods: parsimony, Maximum Likelihood and Bayesian.

Note that one common feature of all phylogenetic inference methods is that they are computationally intensive. Why? If you have three species, there are three possible ways those three species could be related in a fully-resolved tree. If we have ten species there are $34,459,425$ possible trees! If we have 50 species we have $2.75292 \text{»} 10^{76}$ possible trees (if you want to work this out for yourself, the equation used is $N_{trees} = (2n-3)!/2^{n-2} * (n-2)!$; (Felsenstein, 2004)). To put that in context, that's more trees than there are stars in the sky, or grains of sand on Earth. That's a lot of possible trees… This means that no phylogenetic inference method can afford to search all of *tree space*, i.e. the universe of all possible trees. Instead, each method applies number of analytical tricks to avoid this, although we will not go into detail about this here.

2.1.2.1 Parsimony

Parsimony has fallen out of favour as a method of phylogeny building in recent years, but it is still probably the best, and fastest, way to build phylogenies using morphological data. Parsimony works on the basis that the simplest solution is best, so the algorithms select the tree that accounts for the data with the fewest changes in character states across the tree, and outputs the *most parsimonious* phylogeny or phylogenies that satisfy that criterion.

Ideally, this will result in one most parsimonious tree, but often more than one phylogeny meets this criterion. In these cases it is usual to create a *consensus tree*, an average topology across all the most parsimonious trees. There are several ways to construct these, from *strict consensus* trees that include only relationships shown in all most parsimonious trees, to *majority rule consensus*

trees that include all relationships found in at least 50% of the most parsimonious trees. Consensus trees are one way that polytomies can be introduced into phylogenies (see Figure 2.3).

2.1.2.2 Maximum Likelihood (ML)

Maximum Likelihood is a statistical method for estimating the unknown parameters of a model. Likelihood is proportional to the probability of observing the data given the model, *P(data|model)*. The Maximum Likelihood solution is the one that has the highest Likelihood, i.e. the model that best describes the observed data. We will use Maximum Likelihood throughout this book to estimate parameters in many different types of models.

In the case of phylogenetic inference, our model is the tree and an evolutionary model describing how the trait data (e.g. gene sequences or morphological characters) change. We therefore use Likelihood to calculate the probability of the observed trait data given the phylogeny and evolutionary model. The method selects the tree that best accounts for the data, i.e. the phylogeny or phylogenies with the Maximum Likelihood.

Maximum Likelihood is considered to be a better phylogenetic inference method than parsimony for genetic data because we can explicitly define the underlying evolutionary model of DNA evolution, rather than just assuming the smallest number of changes gives the best tree. Remember that DNA is made up of four bases (cytosine [C], guanine [G], adenine [A] and thymine [T]), and these can be split into pyrimidines (C and T) and purines (A and G). Changes from C to T (and vice versa), or A to G (and vice versa), are called *transitions*, and these appear to be easier and thus occur at a higher rate than *transversions*, i.e. changes from C to A (and vice versa), or C to G (and vice versa), or T to A (and vice versa), or T to G (and vice versa). The frequencies of bases can also vary. Multiple models of DNA sequence evolution exist that take into account these various factors. The most commonly used are:

- **JC**. Jukes Cantor (Jukes et al., 1969). This is the simplest model possible. All base changes are equally likely; base frequencies are equal.
- **K2P**. Kimura 2 parameter (Kimura, 1980). Transitions and transversions have different substitution rates; base frequencies are equal.
- **HKY85**. Hasegawa-Kishino-Yano-85 (Hasegawa et al., 1985). Transitions and transversions have different substitution rates; base frequencies can vary.
- **GTR**. General Time Reversible (Lanave et al., 1984). All six types of base change (A to C, A to G, A to T, C to G, C to T, and G to T) occur at a different rate, and base frequencies vary.

Maximum Likelihood approaches can use any of these models, or any other model we might want to define. It is harder to come up with sensible, simple models of evolution for morphological data (although they do exist, for example

the *Mkv* model; (Lewis, 2001)), thus Maximum Likelihood approaches are used less often for morphological datasets.

Maximum Likelihood approaches are very computationally intensive and can be very slow, certainly much slower than parsimony. Like parsimony, often more than one phylogeny has the same Maximum Likelihood so we construct consensus trees.

2.1.2.3 Bayesian

Bayesian methods calculate the probability of the model given the data, *P(model|data)*, i.e. the reverse of Maximum Likelihood. In the case of Bayesian phylogenetic inference, we calculate the probability of the observed phylogeny given the trait data and the underlying evolutionary model. These models are the same as those used in Maximum Likelihood tree inference.

The main difference between Bayesian approaches and both Maximum Likelihood and parsimony, is that we are no longer searching tree space for the "best" tree or trees. Instead, Bayesian methods output a *posterior distribution* of trees. The Bayesian method searches tree space, saving and outputting all trees it finds at regular intervals. This can result in thousands or even millions of possible trees. The user then specifies how many trees they would like to export as a posterior distribution of trees. This will consist of that number of trees, spread across the variety of trees inferred, but with the frequency of their occurrence being based on their probability. For example, if your Bayesian method produces tree topologies A-Z, but topologies A-D are most common, the posterior distribution of trees will mirror that and will contain more trees with topologies A-D, than those with topologies E-Z.

The reasoning behind this is that phylogenetic inference is complex, so the probability of there being just one (or a few) trees that best represent the relationships among species is unlikely. With Bayesian approaches it is possible to get a clearer idea of *phylogenetic uncertainty*, and to actually incorporate this into our understanding of species relationships and downstream comparative analyses.

Because of the way posterior distributions are constructed, consensus trees are not appropriate for Bayesian phylogenies. Instead, we tend to run each analysis on the whole posterior distribution of trees, resulting in a posterior distribution of results. This can be a little more difficult to visualise, but gives a much better understanding of the effect of phylogenetic uncertainty on comparative analyses.

2.1.2.4 Branch support

The trees produced in all three of the methods above may contain measures of *branch support*. These will be shown as numbers above or below the nodes of the phylogeny if you're looking at the published version. Branch support measures tell you how much support there is for a particular branching pattern. There are many ways we can estimate branch support in a phylogeny (Felsenstein, 2004), but the details are not important here. The key thing to remember is that the higher the branch support value, the more confidence we have in that particular branch. Usually these methods give numbers that range from 0-100 (e.g. bootstrap values), but other measures (e.g. Bremer support values) range from 0-1. It should be obvious whether a 0-100 or 0-1 scale is being used from the range of numbers you can see. It's also common for people to collapse branches with low support leading to polytomies.

2.1.3 Where do we get phylogenies from for comparative analyses?

Above we explained how phylogenies are inferred, but you have probably gathered by now that we are not experts in phylogenetic inference! So where do we get phylogenies from for comparative methods? The simple answer is the Internet! Less flippantly, we get them from the published literature. Generally people publishing phylogenies will provide a NEXUS or Newick file containing their phylogeny as part of their supplementary materials. This is often a file ending in `.nex` or `.tre`. In the **online materials** we show you how to read these files into R, and explain more about how these files are structured. If an author has not provided their phylogeny in a format you can read into R (for example they may just have provided a PDF of the topology) you can email them and see if they are willing/able to give you a copy. Some researchers also store phylogenies in online databases such as Treebase[1].

How do you choose the "best" phylogeny for your group? There is no hard or fast rule here, but we do recommend taking some time to think about this carefully. In some taxonomic groups you will have no choice as there may only be one existing phylogeny. In other groups you may be spoiled for choice. How you make the decision will also depend on how much you already know about the group in question. Some questions to ask that may guide you to the best phylogeny include:

- Given your knowledge of the group, does the taxonomy they have used and the characters used to infer the tree make sense? This may be particularly important for morphological trees, but it is also important to check they used an appropriate gene if using molecular trees.

[1] https://www.treebase.org/

- How many of your species are included in the phylogeny? You can add or remove species easily in R (see **online materials**), but if many species are missing it may be better to choose an alternative tree. You can also stitch together multiple trees (see **online materials**).
- What kind of method was used to infer the phylogeny? Does it seem appropriate to you? A phylogeny using genetic data but inferred using parsimony may ring alarm bells…
- Are the branches of the phylogeny well-supported? We didn't talk about branch support above, but you'll often see numbers above the nodes on a phylogeny. These tell you how well supported the relationships are, and generally range from 0-100 or 0-1. Good values will be close to 100 or 1 depending on the method used.
- Are there lots of polytomies or is the phylogeny well-resolved? The methods we use in R can cope with polytomies (see **online materials**) but if there are more resolved phylogenies available these will be better.
- Can you access several different versions of the phylogeny, or a posterior distribution of phylogenies to account for phylogenetic uncertainty?
- If you don't know the group, or phylogenetics, well, don't be afraid to ask the opinion of someone who knows more than you. They may save you hours of pain and indecision!

Inferring phylogenies is really difficult. We make many assumptions and choices at each stage of the process, from data collection to building consensus trees. As such, phylogenies are unlikely to give us the "true" evolutionary history of a group. Instead phylogenies are **hypotheses** about the way evolution happened. It is crucial to remember this when applying and interpreting comparative methods. Even if your comparative method is perfect in every way, you are still basing your analysis on a hypothesis about how species evolved. If this is incorrect, and it almost certainly is (due to missing extinct species etc.), then your results may not be as robust as you imagine.

2.2 Comparative data

2.2.1 Sources of comparative data and data quality

We have talked above about where we get phylogenies from, but what about the comparative data itself? There are a number of options and these will vary depending on your question and the data availability for your study group. Most comparative data will come from a range of sources, including peer-reviewed literature, books, field guides, Internet searches, and curated databases. You can also collect some data from species in the field (e.g. behavioural data) or

from museum specimens (e.g. morphological data). The quality of data from these different sources will vary, so care is needed.

The results of comparative analyses are *only as good as the data you put into them*. Any choices you make in summarising the data need to be recorded and reported in your paper/thesis/report/supplementary material.

2.2.1.1 Peer-reviewed literature, books, field guides, and internet searches.

The easiest way to collect comparative data is to sit in a library and work through all of the books and journals on your study group looking for the data you need. This is often a robust technique, as you can be fairly confident in the legitimacy of these sources, and for some groups these data have not yet been uploaded to the Internet. It is also possible to do all of this using Internet searches instead, but it is important to check the source of any information provided. Journal articles and professional web-pages are likely to be fine, but other web-pages may not always state the source of their information. You should be prepared to make judgment calls about whether you trust a particular source and want to include it in your dataset.

Collecting data manually reveals several issues common to most comparative data collection. It's a sensible idea to decide how you will deal with these issues should they arise.

- Some sources will be more up-to-date than others. Should you record all the data from all sources, or just focus on the most recent?
- Some sources will give actual values whereas others will give approximations. Should you record any data you find, or remove approximations?
- Some data will be based on multiple individuals whereas others will be based on only one.
- Some sources will list values as means, others will list values as maxima or minima. Will you record all of these or choose one?

There are of course many ways to do this, and as long as you **record the decisions you have made** so that someone could repeat the process if necessary, this is not a problem. Our recommendation is to record all data initially, along with some measure of data quality (low, medium or high is sufficient). You can still include this data either by incorporating measures of data quality into your analyses, or by excluding data below a certain data quality threshold when calculating summary data for the species. And don't forget to record the **units** for each measurement. Finally it is extremely important to **record the sources** for each datum you add to your dataset, so that you are able to repeat the process if needed. This also allows you to remove data from any sources you later decide not to trust, and allows you to check individual data points if

there are any issues, for example if one species appears as a large outlier in your comparative analyses this may reflect an error in data collection.

After collecting your data you'll likely need to summarise these data for each species or taxon, i.e. take the mean, median, minimum, maximum etc. It is up to you whether you use means or medians. Often we choose the median because it is less influenced by outliers. The summary data you use and how you calculate it will vary depending on your inputs, for example, if we have maximum values for a variable, we might want to take the overall maximum for the species, but if we have mean values, we might want to take the median for the species (see below). This process may seem trivial, but can have a large influence on your results.

2.2.1.2 Curated databases

Curated databases are popular sources of comparative data. Some will be constantly added to as more data become available e.g. GBIF. Others are static, often because they were released as part of a published paper e.g. Meiri reptile dataset. These sources are popular because you can rapidly download data for a large number of species, making it much faster than going through books and literature manually. But just because a value is published in a database does not mean that it is reliable. The issues noted above still apply, and it is a good idea to check the data and sources of a database in the same way as for books or Internet searches.

2.2.1.3 Practicalities of comparative data collection

How do we practically collect and record comparative data? Let's imagine we have the following three sources of chameleon body length and life history data.

SOURCE 1: Smith et al. 2020.

- *Chamaeleo chamaeleon.* 245 mm, 80 eggs, sexually dimorphic.
- *Brookesia minima.* 33 mm, 2 eggs, sexually dimorphic.
- *Calumma parsonii.* 650 mm, 50 eggs, sexually dimorphic.

SOURCE 2: Jones et al. 2020.

- *Chamaeleo chamaeleon.* Up to 250 cm, Up to 100 eggs.
- *Brookesia minima.* Up to 34 mm, 2 eggs.
- *Calumma parsonii.* Up to 695 mm, Up to 50 eggs.

SOURCE 3: Han et al. 2020.

- *Chamaeleo chamaeleon.* Approximately 10 inches.
- *Brookesia minima.* Approximately 1 inch.

– *Calumma parsonii*. Approximately 24 inches.

We would tend to record these data in a table (probably in Excel or another spreadsheet program) like this.

Species	Measurement	Value	Data_Type	Units	Data_Quality	Approximation	Source
Chamaeleo chamaeleon	length	245	mean	mm	high	FALSE	Jones
Chamaeleo chamaeleon	clutch size	80	mean	eggs	high	FALSE	Jones
Chamaeleo chamaeleon	dimorphic	1	truefalse	NA	high	FALSE	Jones
Brookesia minima	length	33	mean	mm	high	FALSE	Jones
Brookesia minima	clutch size	2	mean	eggs	high	FALSE	Jones
Brookesia minima	dimorphic	1	truefalse	NA	high	FALSE	Jones
Calumma parsonii	length	650	mean	mm	high	FALSE	Jones
Calumma parsonii	clutch size	50	mean	eggs	high	FALSE	Jones
Calumma parsonii	dimorphic	1	truefalse	NA	high	FALSE	Jones
Chamaeleo chamaeleon	length	250	max	mm	high	FALSE	Smith
Chamaeleo chamaeleon	clutch size	100	max	eggs	high	FALSE	Smith
Brookesia minima	length	34	max	mm	high	FALSE	Smith
Brookesia minima	clutch size	2	max	eggs	high	FALSE	Smith
Calumma parsonii	length	695	max	mm	high	FALSE	Smith
Calumma parsonii	clutch size	50	max	eggs	high	FALSE	Smith
Chamaeleo chamaeleon	length	10	mean	inches	medium	TRUE	Han et
Brookesia minima	length	1	mean	inches	medium	TRUE	Han et
Calumma parsonii	length	24	mean	inches	medium	TRUE	Han et

We would also record the full reference for each source in a separate table. Recording the data like this makes it easy to enter in Excel, and also makes it easy for us to exclude certain types of data or sources should we decide we don't trust them enough to include in our analyses.

Note that this will not be the ideal format for using the data in R, so you will need to manipulate or *wrangle* the data first. For most analyses, data need to be in a *tidy* format, meaning:

– Each variable must have its own column.
– Each observation must have its own row.
– Each value must have its own cell.

In R, the `tidyr` and `dplyr` packages are useful for doing this. Or you could use something like Excel if you are less confident with R. Make sure to keep the raw data too in case you accidentally introduce any errors, and so that you can repeat the data collection and analyses if needed. See https://r4ds.had.co.nz/tidy-data.html for more details on what tidy data is and how to do this.

To wrangle this data, we would first convert the units to a standard unit for

each measurement. Here we have length in mm and inches, so we would convert the inches into mm. The three altered rows are shown below. Note that converting from inches to mm gives us the false impression of precision, i.e. two values are now 25.4 and 609.6, rather than whole numbers making it look like these were measured accurately rather than approximated. It's worth looking out for this in other data, especially curated databases where this kind of conversion is typical.

Species	Measurement	Value	Data_Type	Data_Quality	Approximation
Chamaeleo chamaeleon	length	254.0	mean	medium	TRUE
Brookesia minima	length	25.4	mean	medium	TRUE
Calumma parsonii	length	609.6	mean	medium	TRUE

Next we might decide to exclude low quality records, and remove any approximations. We might need to do something more complicated if we only have one record for a species and it is an approximation. In this case we are going to keep the approximations, and use all the data. But you should choose what is sensible for your dataset.

Finally, we want to summarise the data for each species, to get means/medians and possibly also minima and maxima. It is up to you whether you use means or medians. Often we choose the median because it is less influenced by outliers. The summary data you use and how you calculate it will vary depending on your inputs for example here we might want to extract:

- *Median of mean length values.* This will tell us about the overall **mean** value for length across all sources.
- *Maximum of maximum length values.* This will tell us about the overall **maximum** value for length across all sources.
- *Median of mean clutch size values.* This will tell us about the overall **mean** value for clutch size across all sources.
- *Maximum of maximum clutch size values.* This will tell us about the overall **maximum** value for clutch size across all sources.
- *Median dimorphism score.* This will tell use whether the species is predominantly considered to be dimorphic or not.

Our final dataset for analyses might look something like this.

Species	median_length	max_length	median_clutchsize	max_clutchsize	dim
Brookesia minima	29.2	34	2	2	NA
Calumma parsonii	629.8	695	50	50	NA
Chamaeleo chamaeleon	249.5	250	80	100	NA

Note that each variable is now its own column. This will help us when using the data in analysis programs like R. R code to achieve the above can be found in the *online materials*.

[CHAPTER ENDING MATERIAL STYLE STILL TBC BUT WILL BE CONSISTENT ACROSS THE BOOK]

2.3 Summary

2.4 Online stuff

2.5 MCQs

Chapter 3

Foundations 2: Introduction to models of trait evolution and phylogenetic signal

[CHAPTER OPENING STYLE STILL TBC AND WILL BE CONSISTENT ACROSS CHAPTERS. CURRENT CONTENT IS PLACEHOLDER ONLY]

The previous chapter introduced you to the basic building blocks of comparative methods; phylogenies and trait data. We are now going to shift gears and think about some of the theory and models underpinning comparative methods.

At the very simplest level, comparative methods are interested in how traits are distributed across trees, and how these traits change through time. Are there clusters in the tree where traits of the species at the tips are similar or identical? Or are traits spread evenly across the tree? How quickly do trait values change through time? etc.

The models we use to describe how traits change will vary depending on whether our traits are continuous (e.g. body size, leaf area) or discrete (e.g. presence or absence of a tail, growth form, number of legs).

This chapter will introduce you to the basic ways we work with traits and trees. It will cover:

- Modelling continuous traits.
  - the Brownian Motion (BM) model of evolution.
- Modelling discrete traits.
  - ARD, ER, SYM
- Phylogenetic signal

3.1 Modelling continuous traits - the Brownian Motion model of evolution

For continuous traits, the most commonly used model of trait evolution is the *Brownian motion model*, and many more complex models have been built upon it. Therefore we will spend some time discussing it here. If you're already familiar with the Brownian model you might want to skip to the summary at the end of the section just to check that you recall all the key features of the model. Otherwise, join me on an exploration of imaginary cats!

3.1.1 An introduction to Brownian Motion

The Brownian model or BM model (Cavalli-Sforza and Edwards, 1967; Felsenstein, 1973) is assumed to be the underlying mode of evolution in the majority of phylogenetic comparative methods (though this assumption is rarely tested; Freckleton and Harvey 2006). It can be fully described using just two parameters $z(0)$ and σ^2.

$z(0)$ is the starting value of the trait we are interested in. This is the mean value for the trait seen in the ancestral population before there have been any changes to the trait. σ^2 is the rate at which evolution (i.e. trait change) occurs. Some of you might already be wondering why we use σ^2 here, as σ^2 is usually used to signify *variance*. And you'd be right, it is a variance! But how and why?

Here's the explanation. At the core of Brownian motion is the normal distribution (Figure 3.1). In the Brownian motion model of evolution, the amount that a trait changes over any time interval is drawn at random from a normal distribution with a **mean of 0** and a **variance of** σ^2. Because the mean is 0, we can draw positive numbers causing the trait to increase, we can draw negative numbers causing the trait to decrease, and we expect the mean change in the trait over time to also be 0. How large these changes are depends on the value of σ^2. In Figure 3.1 you can see that as σ^2 gets larger, so does the spread of the normal distribution, meaning that large values are more likely to get drawn from the distribution as σ^2 increases, leading to larger increases or decreases in the value of the trait. This means evolution will be faster when σ^2 is larger. Because of this, σ^2 represents the *rate of evolution*, and is referred to as the *Brownian rate parameter*.

Because the direction of change in trait values at each step is random, Brownian motion is often described as a "random walk".

Let's run a quick example simulation to see what is going on. Our trait can be any continuous feature of an organism, but for simplicity let's use body mass in (imaginary!) cats. First let's run one simulation to show how the body mass of one cat lineage changes through time under the Brownian model.

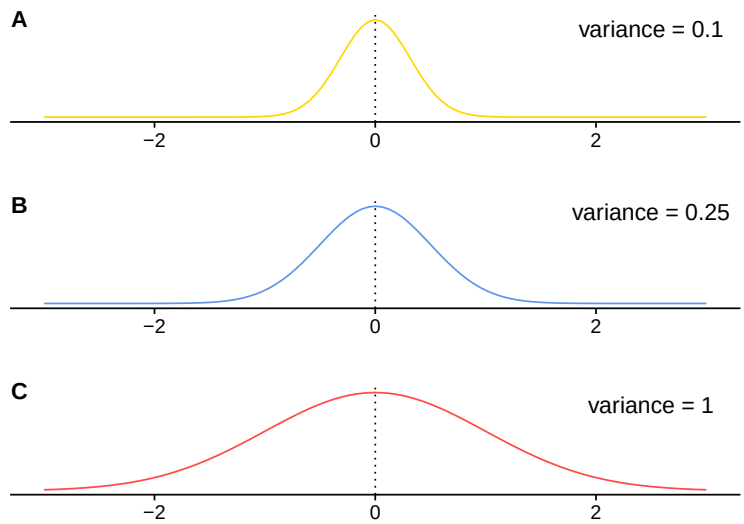

Figure 3.1: Three normal distributions with the same mean (0) but different variances.

1. At the start, the trait value is $z(0)$. Here we'll start at $time = 0$ with an ancestral cat that weighs 4 kg. The rate of evolution, or Brownian rate parameter, σ^2, is 0.25 (like the normal distribution in Figure 3.1B).

2. To determine the value for body mass at $time = 1$, we draw a random number from the normal distribution with $mean = 0$ and $\sigma^2 = 0.25$. In this example the value is -0.1533193. So the body mass of a cat at $time = 1$ is $4 + -0.1533193 = 3.84668$.

3. To determine the value for body mass at $time = 2$, we draw a random number from the normal distribution with $mean = 0$ and $\sigma^2 = 0.25$. In this example the value is -0.89065422. So the body mass of a cat at $time = 2$ is $3.84668 + -0.89065422 = 2.95603$, recall that 3.84668 is the body mass at $time = 1$.

4. If we continue this until we get to $time = 10$ we get a graph that shows the evolution of the trait through time that looks like Figure 3.2 (the numbers are the body mass values, rounded to two decimal places).

A few things to note. First, the direction of change is random, there is no trend towards increasing or decreasing body mass. Second, the changes do not depend on the changes that happened previously, i.e. an increase in body mass at time 1 does not mean an increase (or decrease) at time 2 is more or less likely. The random nature of trait changes is why the Brownian model is often referred to as a **random walk**. We sometimes describe it as being like a drunk person walking down the road, wobbling about but in no particular direction…

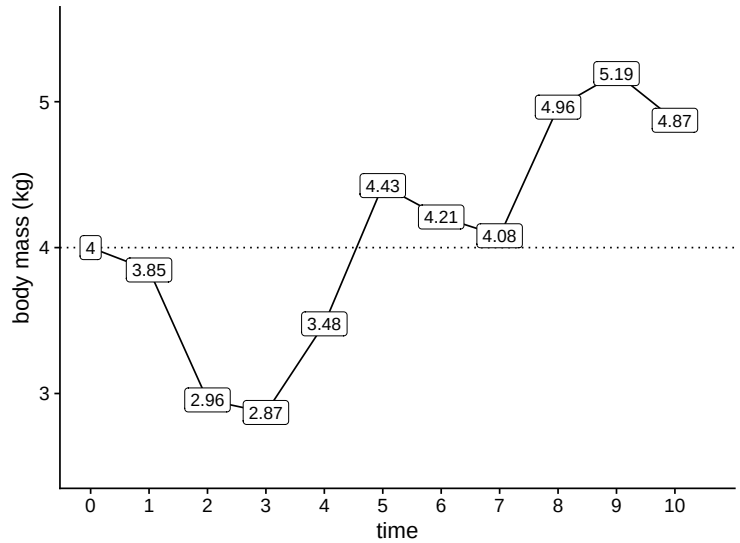

Figure 3.2: A simple simulation showing how the body mass of cats changes through time under a Brownian model with $z(0) = 4kg$ and $sigma^2 = 0.25$.

Although our examples here use t = 1, t = 2 etc. to illustrate what is happening, Brownian motion is a continuous time process, and so time does not have discrete steps in the model.

The example above includes only one simulation to give you a basic idea of what is going on. To get a better idea of the patterns we expect to see, let's run 100 simulations instead, over a longer time period. Because the Brownian model is a random walk, we expect each simulation to be slightly different depending on the values drawn out of the normal distribution each time.

As above, we will start at $time = 0$ with an ancestral cat that weighs 4 kg (i.e. $z(0) = 4$), and use σ^2 of 0.25 as the rate of evolution (Figure 3.3A).

Notice how the variance in body mass (i.e. the spread of different values of body mass) *increases linearly with time* (it's not perfectly linear here, but it would be if we ran more simulations; Figure 3.3B). Meaning that body mass variance is greatest over the longest time intervals. In biological terms this means we expect to see more variance in the trait values of groups that have been evolving for longer.

If we vary the value of σ^2 we can look at the effect of changing the rate of evolution on the values of body mass we evolve in our cat simulations. If the rate of evolution is lower, then variance in body mass accumulates more slowly (Figure 3.4A); if the rate of evolution is higher, then variance in body mass accumulates more quickly (Figure 3.4C). The overall variance of the Brownian

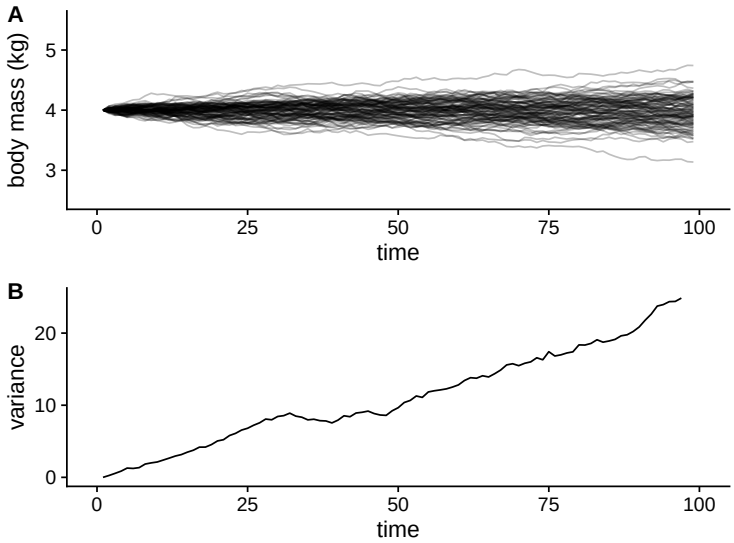

Figure 3.3: 100 simulations showing how the body mass of cats changes through time under a Brownian model with $z(0) = 4kg$ and $sigma^2 = 0.25$ (A); and a plot showing how variance in body mass across the simulations increases linearly through time (B).

process is equal to the rate of evolution multiplied by the amount of time that has elapsed.

```
## Warning: Removed 5 row(s) containing missing values (geom_path).
```

One final feature to highlight is what the body mass values look like at the end of the simulations, i.e. where $t = 100$. If we assume that the present-day is $t = 100$, we can imagine that these values are equivalent to the body mass values for the species at the tips of our phylogeny.

Because trait changes are drawn from a normal distribution with a mean of 0, on average we expect no change in body mass through time. So if we run enough simulations, the mean of the trait values at the end of the simulations should be equal to the starting value $z(0)$, in this case 4 kg. The rest of the values should be evenly distributed around the mean, so we expect the body mass values to be normally distributed.

As demonstrated above, the variance of these normal distributions of body masses will depend on the rate of evolution (low rates result in smaller variances; high rates result in larger variances; Figure 3.5) *and* the time elapsed (shorter times result in smaller variances; greater times result in larger variances; Figure 3.3). More precisely, we can define the overall variance of the Brownian process as the rate of evolution multiplied by the amount of time that has elapsed.

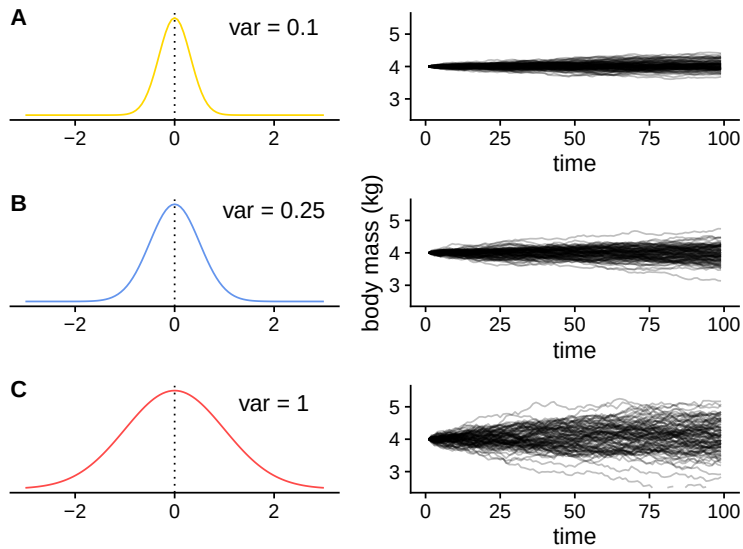

Figure 3.4: 100 simulations showing how the body mass of cats changes through time under a Brownian model with $z(0) = 4kg$ and (A) $sigma^2 = 0.1$; (B) $sigma^2 = 0.25$; (C) $sigma^2 = 1$.

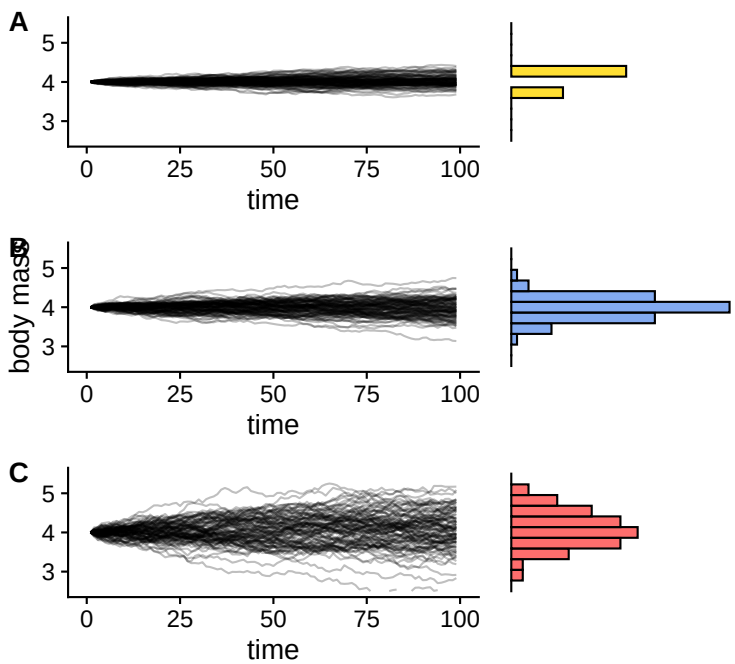

Figure 3.5: Three Brownian motion simulations (left) with different values of sigma (A = 0.1, B = 0.25, C = 1) and the resulting trait distributions at the end of the simulation (right).

The examples above ignore the topology of the phylogeny. What if we want to model the evolution of all species in a phylogeny? This is usually how we use the Brownian model, and it is very simple to extend. At each time point where there is speciation event, we simply start a new Brownian simulation for the two new species. Let's do this with the cat example, assuming there are speciation events at time 3, 5, and 7, creating a phylogeny like the one in Figure 3.6.

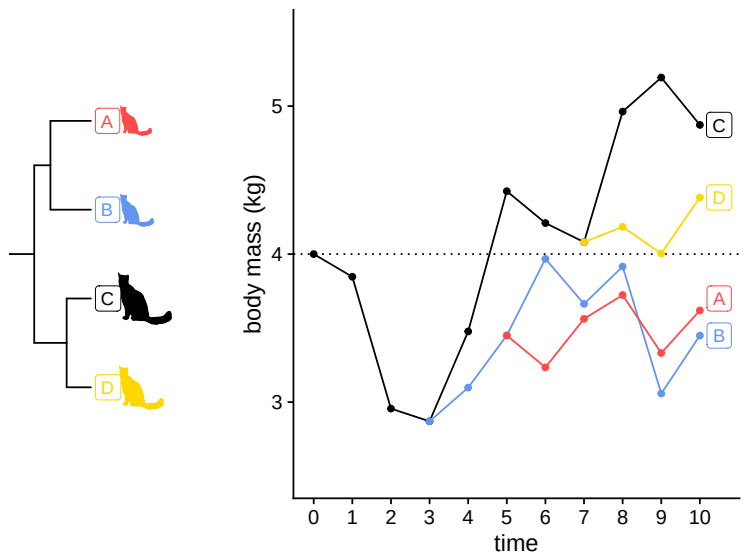

Figure 3.6: A simulation showing how the body mass of four lineages of cats changes through time under a Brownian model with $z(0) = 4kg$ and $sigma^2 = 0.25$ (A), and the phylogeny of the cats (B). The size of the cats indicates the final body size of the lineage in the simulation.

A few things to note here.

First, when a lineage speciates, the new lineages start at the same trait value, and over time they can evolve away from each other. This means that close relatives will be more similar in their trait values than more distant relatives, and differences in trait values between species will depend on how long ago they diverged from a common ancestor. For example, here we expect cat A and cat B to be more similar to one another than they are to cat C and cat D, and that cat A and cat B should be more similar to one another than cat C and cat D are to one another, because cat A and cat B diverged more recently than cat C and cat D (Figure 3.6). More precisely, the correlation among trait values is proportional to the extent of shared ancestry for pairs of species.

Second, we used the same normal distribution throughout to randomly draw our trait change values. This means that σ^2 was also the same throughout. This means that the *rate of evolution across the tree is constant*, i.e. it is the same for all the lineages. This is a big assumption of the Brownian model, and it is

often broken in real life scenarios (for example the rapid evolution of clades in an adaptive radiation). We will return to this later in the book.

To summarise the above, the key features of the Brownian motion model are:

- Traits evolve at random, i.e. by a random walk.
- The direction and magnitude of the trait change at one time interval is independent of the direction and magnitude of the trait changes at other time intervals.
- There is no overall drift in the direction of evolution.
- Traits evolve at a rate σ^2.
- The rate of evolution is constant.
- Variance in traits increase (linearly) in proportion to time.
- The overall variance of the Brownian process is the rate of evolution times the amount of time that has elapsed.

In terms of the model parameters:

- The model can be fully described using just two parameters $z(0)$ and σ^2.
- $z(0)$ is the starting value of the trait we are interested in. This is the mean value for the trait seen in the ancestral population before there have been any changes to the trait.
- σ^2 is the rate at which evolution (i.e. trait change) occurs. It is also referred to as the Brownian rate parameter.

Some biological implications of the model are:

- Correlation among trait values is proportional to the extent of shared ancestry for pairs of species, i.e. close relatives are more similar than more distant relatives.
- Different lineages evolve at the same rate.

What if evolution is non Brownian? You've probably already realised that evolution is not always going to be Brownian. There are a number of other models of evolution, and we will return to these in *Chapter 7*.

3.2 Modelling discrete traits - simple models

Many traits we are interested in studying are not continuous traits, instead they can be lumped into distinct categories, so we refer to them as discrete traits. Discrete traits have a number of states, and comparative methods can be used to investigate changes between states across a phylogeny. For example, if we are interested in the evolution of flightlessness in birds, we might want to know

how many times this has evolved across the phylogeny, whether flying birds are more likely to evolve to be flightless, or whether flightless birds are more likely to evolve to fly, and whether flightless birds can ever evolve to fly again once they have lost the ability.

In the flightlessness in birds example, we have two states: flying and flightless. We could also have intermediate states, e.g. semi-flightless. Traits can be ordered or unordered. Ordered traits have a set order of state changes, for example flying to semi-flightless to flightless. Unordered traits can change in any way. Traits can also be reversible or non-reversible, for example, if flightless bird lineages can never go on to evolve flight again, the trait is irreversible. The most simple way to model discrete traits is to assume that changes are unordered and reversible, so we will stick to this here.

We've actually already met some models of discrete trait evolution over trees. Back when we discussed phylogenetic inference we mentioned several models of DNA sequence evolution, including the Jukes Cantor (Jukes et al., 1969), and General Time Reversible (GTR) (Lanave et al., 1984) models. The most basic models for discrete character evolution are analogues of these models.

3.2.1 Mk/Equal Rates model (ER)

The simplest model for discrete trait evolution is a direct analogue of the Jukes Cantor model. Except instead of changes in base pairs, we model changes in character states. We call this the Mk model ((Lewis, 2001, Pagel (1994))). The M stands for Markov, because the Mk model assumes states change according to a Markov process, i.e. the probability of changing from one state to another is independent of what has happened previously, it only depends on the current state of the trait. The k is there because we generally use k to represent the number of parameters in a model. In models of discrete trait evolution, k is the number of states. For our flying/flightless example $k = 2$, for the flying/semi-flightless/flightless example $k = 3$.

In the Mk model, all trait changes are equally likely, and all occur at the same rate. Because of this we often call the Mk model the Equal Rates or ER model.

We can visualize the Q (rate) matrix for an Mk model of our flightlessness example like this:

-	flying	semi-flightless	flightless
flying	-	1	1
semi-flightless	1	-	1
flightless	1	1	-

The off-diagonals are the transition rates from state 1 to 2, 1 to 3, 2 to 1 and

so on (read rows then columns). We typically designate individual transition rates in the form $q_i j$, which means the rate of going from state i to state j. Here, the 1 in all off-diagonal elements represents the fact that rates are the same regardless of what state i and j are, and the direction of that change. The diagonal elements $q_i i$ give the rate of not changing and are computed as the negative sum of the non-diagonal row elements. This is so that the rows sum to zero. Here we've just left this as "-".

3.2.2 Symmetric rates model (SYM)

The Mk model assumes changes between states are equally likely, but we have good reasons for thinking this might not be the case in many biological systems. Some changes in states are likely to be much easier than others, for example losing complex traits like eyes or limbs is likely to be easier than gaining them. The Mk model can be modified to allow for these more complex situations. The Symmetric rates or SYM model (Paradis et al., 2004) is one such modification where the rate of change can differ among pairs of states, but within each pair of states the rate of change is the same in both directions. The number of different rates is three for $k = 3$.

Using our flying/semi-flightless/flightless example, this means that the rate of change from flying to semi-flightless is exactly the same as the rate of change from semi-flightless to flying. However, these rates differ from the rate of change from flying to flightless (and vice versa), and also differ from the rate of change from semi-flightless to flightless (and vice versa). In terms of the DNA evolution models this is analagous to the GTR model (Lanave et al., 1984).

The Q matrix for our flying/semi-flightless/flightless example looks like this:

-	flying	semi-flightless	flightless
flying	-	1	2
semi-flightless	1	-	3
flightless	2	3	-

It's worth noting that if you only have two states, i.e. $k = 2$, then the Symmetric model is exactly the same as the Mk/Equal Rates model.

-	flying	flightless
flying	-	1
flightless	1	-

3.2.3 All Rates Different (ARD) model

The final modification of the Mk model we're going to cover here is the All Rates Different or ARD model (Paradis et al., 2004). In this model not only does the rate of change differ among pairs of states, it can also differ within each pair of states depending on the direction of the change. For k states, we get $k^2 - k$ rates, which can be a pretty large number as k increases.

The Q matrix for our flying/semi-flightless/flightless example looks like this:

-	flying	semi-flightless	flightless
flying	-	1	2
semi-flightless	3	-	4
flightless	5	6	-

Biologically, this is a useful model to fit if you think some of your states are not reversible, or very difficult to reverse, for example if you think the rate of evolution from flightless to flying is likely to be close to 0. It often works well for big complex datasets that span a variety of different states. For example, if you had a tree of all vertebrates and wanted to examine transition rates among different dietary strategies, this model would be worth examining.

You may have noticed that there is no analaogous model to ARD in DNA evolution. Molecular biologists don't use this kind of model because of potential *over-fitting* (see below), which is also a very important point to consider when chosing whether or not to fit an ARD model . It's also really important for some of the other methods we'll use later in the book, so I'll explain what I mean by over-fitting in the box below. If you're fitting an ARD model with six rates to a phylogeny containing only 30 species, you definitely need to stop and think about over-fitting.

Over-fitting occurs where a model has more parameters than can be justified by the data. It can mean the model looks like it beautifully fits the data, but if you were to add additional data it would be useless.

Imagine you have data on the swimming speed of ten penguins (let's call them Pingu, Pinga, Misha, Wheezy, Chilly Willy, Mumble, Skipper, Kowalski, Rico, Private and Feathers McGraw). You could fit a model (we could use a linear regression here) to see if body mass is correlated with swimming speed, and get a regression with an r^2 of 0.5. This suggests around half of the variance in swimming speed is explained by body mass. The other half might be explained by all kinds of things, including specific attributes of the particular penguins or errors in your measurements. We haven't fully explained variation in swimming speed, but it's still a useful model because if we weighed another penguin we'd have a reasonable

chance of estimating roughly what its swimming speed might be from its body mass.

If we wanted to increase the variance explained by the model as much as possible, we can start adding more variables. For example, maybe the faster penguins also have longer flippers so we add in flipper length. This might be sensible, and if we had more penguins in our dataset this would be fine. But what if we also add in a variable that accounts for Wheezy being slow due to his asthma, and another that accounts for Feathers McGraw being faster because he's evil? If we add enough variables, eventually we get to a model with an r^2 of 1. This means we can perfectly explain the swimming speed of our ten penguins, but we've got no hope of applying it to any new penguins we find. This is over-fitting, and the model not useful. Visually you can think of it the difference between fitting one line of best fit to a regression of X against Y, compared to joining all the points up.

There's no hard and fast rule about how much data you need to add another variable to your model, but some rules of thumb range from 20-30 data points for each variable. Complex models are always likely to fit data better just because they have more parameters, not necessarily because they mean anything biologically meaningful. You can avoid this problem by thinking very carefully about the models you chose to fit, why you think they are sensible, what you predict you will find, and how you will interpret the results, *before* fitting the model.

3.2.4 A quick note about ancestral state estimation

You often hear about using phylogenies for *ancestral state reconstruction*, along with suggestions that these reconstructions can tell us what the ancestor of a group looked like, how it behaved, what it ate for breakfast etc. Sadly this is not as magical as it seems, and we should refer to this as ancestral state *estimation* NOT *reconstruction*. Ancestral state estimation is probably one of the most over-used and uninformative methods in the phylogenetic comparative methods toolkit.

There are many reasons to be highly skeptical of ancestral state estimates and interpretations of macroevolutionary patterns and process that are based on them, not least because we know that phylogenies are only *hypotheses* and will often miss out large chunks of extinct diversity. Ancestral state estimation methods are based on the models we discussed above, meaning that your ancestral states are at best weighted averages of your trait based on branch lengths and the model used. How much trust you would want to put in such a result is up to you, but hopefully you now know enough to be very skeptical! However,

certain phylogenetic comparative methods require that you define the state of the ancestor for them to work. We will return to this later in the book.

Ancestral state estimates (**not** reconstructions) are, at best, weighted averages of your trait based on branch lengths of the phylogeny and the model of evolution used. If your phylogeny or traits or model have error (and they always will) then your ancestral state estimates should be interpretted with care.

3.3 Phylogenetic variance-covariance matrices

A useful analytical tool that we will use in a number of methods in this *Primer* is to represent a phylogeny as a *phylogenetic variance–covariance matrix*. First let's think about what we mean by variance and covariance in the context of traits on phylogenies. To do this we need to go back to the Brownian motion model.

Remember that under Brownian motion, variance in a trait increases linearly in proportion to time. So species that diverged recently will have similar trait values, and species that diverged a long time ago will have very different trait values. In more technical terms we can say that the expected *covariance* between species' trait values at the tips of the phylogeny is exactly proportional to the shared history of the species involved, i.e. the sum of their shared branch lengths. In addition, the expected *variance* of a trait value for a given species is proportional to total length of the tree, i.e. the summed branch length from the root to the tip for that species [19], also refered to as the *root-to-tip* distance.

Figure 3.7 shows how we get the root-to-tip distances from a phylogeny. The letters a-f represent the branch lengths. Because this is an *ultrametric* tree, the root-to-tip distance for each species will be the same, i.e. the same amount of time has elapsed. So in the example in Figure 3.7, a *and* b + c *and* b + d + e *and* b + d + f all add up to the same number. Likewise, branch c is the same length as branches d + e, and branch e is the same length as branch f. Take a good look at the phylogeny in Figure 3.7 to make sure you understand this.

As already mentioned above, these root-to-tip values give us the variance of the trait across the tree. But what about the covariances among species, i.e. the shared branch lengths? We can extract these in the same way. The shared branch length between the robin and the deer is simply b + d (note that we don't include e or f because these branches only lead to either the robin *or* the deer). The shared branch length between the robin and the spider is b, because this is the only branch that leads towards them both. Likewise, the shared branch length between the deer and the spider is also b. The slightly tricky one is what is the shared branch length between the robin/deer/spider and the

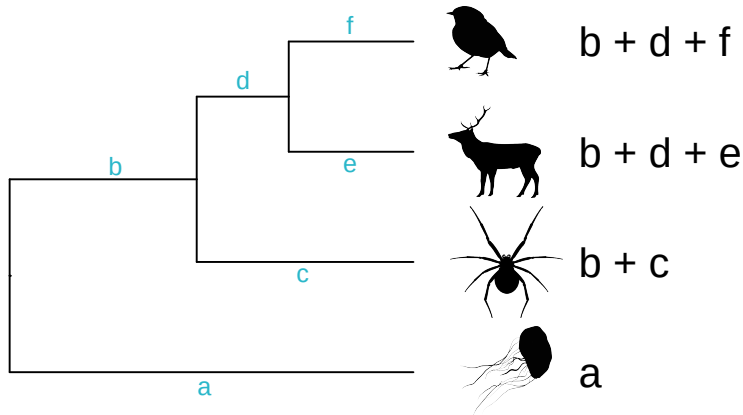

Figure 3.7: Phylogeny showing branch lengths and root-to-tip distances for each tip taxon

jellyfish? They don't share a branch at all in this phylogeny so we say their shared branch length is zero.

Why spend this time thinking about the trait variances and covariances? Because we know these values we can represent the phylogeny as a *phylogenetic variance–covariance matrix*. A phylogenetic variance–covariance matrix is an n x n matrix, where n is the number of species in the phylogeny, and the diagonals are the variances, and the off-diagonals are the covariances. For our example in Figure 3.7 the phylogenetic variance–covariance matrix looks like this.

-|robin|deer|spider|jellyfish| robin|b+d+f|b+d|b|0| deer|b+d|b+d+e|b|0| spider|b|b|b+c|0| jellyfish|0|0|0|a|

We will use phylogenetic variance–covariance matrices at various points in this *Primer*, including in the next section on phylogenetic signal. Often people mention these matrices (sometimes they will be referred to as phylogenetic vcvs) without explaining what they are, but hopefully the outline above has made them seem a bit less opaque and scary!

As you may already be aware, variance–covariance matrices are used in many statistical methods, not just phylogenetic comparative methods. The fact that we can represent a phylogeny in this way will make our lives way easier in the next few chapters!

3.4 Phylogenetic signal

Phylogenetic signal is one of the most commonly mentioned concepts in phylogenetic comparative methods. Unfortunately there's a lot of misunderstanding about what it is and what it means! Below we summarise what it is, how we can measure it, and how we can (and can't) interpret the results.

3.4.1 What is phylogenetic signal?

Phylogenetic signal is the pattern where close relatives are more similar than distant relatives (Harvey and Pagel, 1991). Species' traits can show high or low phylogenetic signal. If phylogenetic signal is high, this means that closely related species have similar trait values, and that trait similarity decreases with phylogenetic distance. If phylogenetic signal is low, then trait values might vary randomly across a phylogeny, with distant relatives sharing similar trait values, and close relatives having different trait values (Kamilar and Cooper, 2013).

You may see papers that use different terms to refer to phylogenetic signal, including phylogenetic effects, phylogenetic constraints and phylogenetic inertia. Unfortunately these terms are poorly defined and mean different things to different people. We therefore recommend that these terms are avoided.

3.4.2 How do we estimate phylogenetic signal?

Like many of the methods we explore in this *Primer*, there are many different ways to estimate phylogenetic signal, each with their own strengths and weaknesses (see reviews by see Munkemuller et al. (Münkemüller et al., 2012) and (Hardy and Pavoine, 2012)). Here, we'll just focus on the two most commonly used metrics: Pagel's λ (Freckleton et al., 2002, Pagel (1999),Pagel (1997)) and Blomberg's K (Blomberg et al., 2003). These metrics are for continuous characters only, so we also include one measure for binary data (Fritz and Purvis, 2010), though this is less widely used. Practical examples of how to estimate these in R are provided in the *online supplement*.

Both λ and K use an explicit model of trait evolution - our old friend the Brownian motion model. If you've forgotten how this model works it might be worth re-reading this section before you begin.

3.4.2.1 Pagel's λ

Pagel's λ was introduced by Pagel (Pagel, 1997, Pagel (1999)) in the early 90s and is probably the most commonly used measure of phylogenetic signal. This

is in part because λ is not just a measure of phylogenetic signal, it is also often used to transform the branch lengths of a phylogeny for use in other analyses; we return to this in *Chapter 4*.

λ transforms the branches of the phylogeny so that it best explains trait variation among species at the tips. For example, if species are less similar in their trait values than expected given their shared branch lengths, λ transforms the tree so the branches are shorter. In practical terms, λ transforms the *off-diagonal values*, or the covariances between pairs of species, of a phylogenetic variance–covariance matrix (see the section above). Remember that the covariances in a phylogenetic variance–covariance matrix are equal to the sum of the shared branch lengths of the species. In terms of the phylogeny, these covariances represent the internal branches of the tree. Importantly, λ does not transform the variance of the tree, which we learned in the section above is equal to the root-to-tip distance or the total length of the tree. This means that the total length of the tree stays constant.

To make this clearer let's use our example from Figure 3.7 but let's add some numbers for the branch lengths (Figure 3.8).

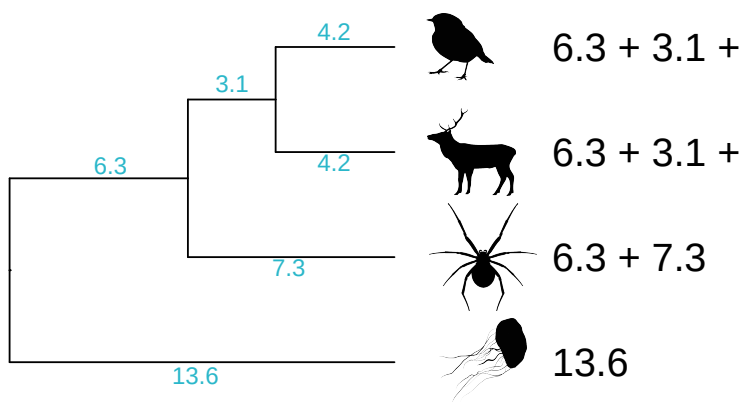

Figure 3.8: Phylogeny showing branch lengths. The root-to-tip distances for each tip taxon are 13.6

Note that the root-to-tip distance for each species sums to 13.6. This is the variance, or diagonal of our phylogenetic variance–covariance matrix. The covariances, or off-diagonals, are the shared branch lengths between pairs of species. The shared branch length between the robin and the deer is 6.3 + 3.1 = 9.4. The shared branch length between the robin and the spider (and the deer and the spider) is 6.3. The shared branch length between the robin/deer/spider and the jellyfish is zero (see section above if this is not clear). Therefore, the phylogenetic variance–covariance matrix for this phylogeny is:

	robin	deer	spider	jellyfish
robin	13.6	9.4	6.3	0.0
deer	9.4	13.6	6.3	0.0
spider	6.3	6.3	13.6	0.0
jellyfish	0.0	0.0	0.0	13.6

λ can be any number between 0 and 1. λ of 0 indicates that there is no phylogenetic signal in the trait, i.e. that the trait has evolved independently of phylogeny and thus close relatives are not more similar on average than distant relatives. If λ is 1 this indicates that there is strong phylogenetic signal, and the trait has evolved according to the Brownian motion model of evolution.

What do λ values of 0, 1 and anything in between, do to the phylogenetic variance–covariance matrix and the phylogeny?

When λ is 0, all of the off-diagonal elements of the phylogenetic variance–covariance matrix are also 0 (any number multiplied by zero is equal to zero). In our example above (Figure 3.8) the phylogenetic variance–covariance matrix will look like this:

	robin	deer	spider	jellyfish
robin	13.6	0.0	0.0	0.0
deer	0.0	13.6	0.0	0.0
spider	0.0	0.0	13.6	0.0
jellyfish	0.0	0.0	0.0	13.6

This means all the internal branches now have a length of zero. This produces a tree which is just one big polytomy, also refered to as a *star phylogeny* (Figure 3.9).

When λ is 1, all of the off-diagonal elements of the phylogenetic variance–covariance matrix are exactly the same (any number multiplied by 1 remains unchanged). In our example above (Figure 3.8) the phylogenetic variance–covariance matrix will look like this, i.e. the same as before the λ transformation. The tree also stays the same as in Figure 3.8.

	robin	deer	spider	jellyfish
robin	NaN	0	0	0
deer	0	NaN	0	0
spider	0	0	NaN	0
jellyfish	0	0	0	NaN

Many explanations of Pagel's λ state that the maximum value of λ is 1. Actually, it can be bigger than 1 if close relatives are *more* similar than expected under the Brownian motion model of evolution. However, most tools that allow you to estimate λ restrict it to a maximum of 1 because

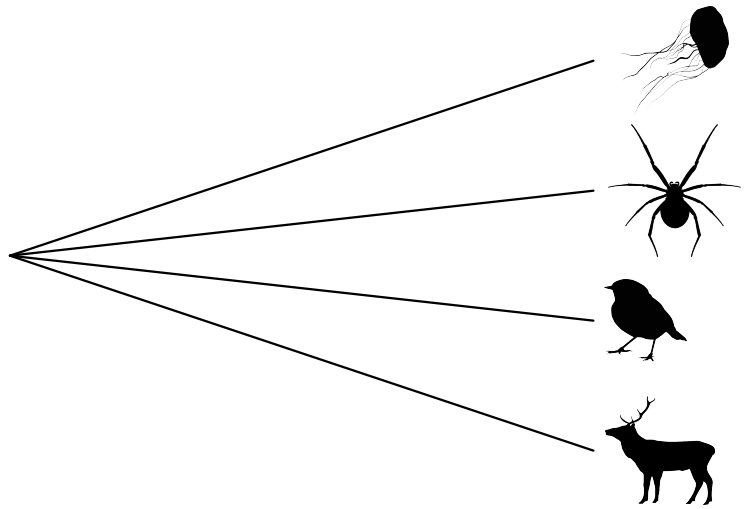

Figure 3.9: Star phylogeny

> covariances cannot exceed variances in a phylogenetic variance–covariance matrix, and this is what would happen if λ was bigger than 1.

Values of λ between 0 and 1 indicate that although there is phylogenetic signal in the trait, it has evolved according to a process other than pure Brownian motion (Freckleton et al., 2002). We'll discuss how to interpret phylogenetic signal in more detail later in this section. But to show you how this affects the phylogenetic variance–covariance matrix and the tree, let's assume a λ of 0.5. This will multiply all the off-diagonal elements of the phylogenetic variance–covariance matrix by 0.5 resulting in a tree as shown in Figure 3.10, and a matrix like this:

	robin	deer	spider	jellyfish
robin	13.60	4.70	3.15	0.0
deer	4.70	13.60	3.15	0.0
spider	3.15	3.15	13.60	0.0
jellyfish	0.00	0.00	0.00	13.6

Understanding what has happened to the branch lengths in Figure 3.10 is a little tricker. Remember that λ only acts on the off-diagonals/covariances of the matrix. These are the *internal* branches of the phlyogeny, i.e. not the branches that lead to the tips. If λ is less than 1, this means that those branches get shorter. Here The internal branch lengths have all been halved because $\lambda = 0.5$.

However, we also know that λ does not act on the diagonal/variances of the matrix that represents the total tree length. If total tree length does not change, but internal branches get shorter, then the branches leading to the tips must get

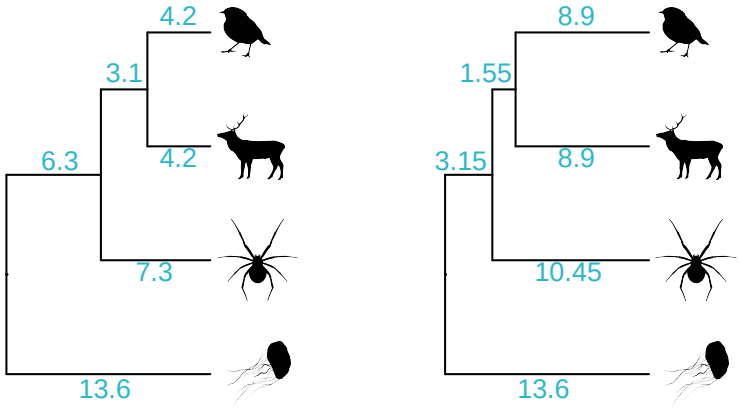

Figure 3.10: Phylogeny showing branch lengths in the original tree and a transformed tree where lambda = 0.5. The root-to-tip distances for each tip taxon are 13.6

longer. Here to ensure the root-to-tip distance for each species is still 13.6, the branches leading to the robin/deer need to be 8.9 (13.6 - 4.7), and the branch leading to the spider needs to be 10.45 (13.6 - 3.15).

We generally estimate λ using Maximum Likelihood. This means that for a given trait and phylogeny, we search for the value of λ that best explains trait variation among species at the tips of the phylogeny, i.e. the Maximum Likelihood value of λ. We can test if λ is significantly different from zero (i.e. no phylogenetic signal) or 1 (i.e. the Brownian motion model expectation) using likelihood ratio tests comparing a model with the observed Maximum Likelihood value of λ to a model with a fixed λ of zero or 1. Some methods that estimate λ will do this for us automatically (e.g. in the `pgls` function in the R package `caper` (Orme et al., 2013); see *online supplement*).

3.4.3 Blomberg's *K*

Rather than transforming the tree like λ, Blomberg's K instead measures phylogenetic signal by quantifying the amount of observed trait variance relative to the trait variance expected under Brownian motion (Blomberg et al., 2003). It's a little tricky to explain how K is estimated, especially in advance of *Chapter 4* which introduces you to phylogenetic generalised least squares (PGLS) models. However, you should be able to understand the general concepts without needing to know the mathematical details. We will first talk about K in very general terms, and then explain how it is estimated in more detail. You don't need to understand the maths to use K in your work, but it's good practice to try.

The equation for Blomberg's K is:

$$K = \frac{observed \frac{MSE_0}{MSE}}{expected \frac{MSE_0}{MSE}} \tag{3.1}$$

The important part of the equation to think about initially is the $observed \frac{MSE_0}{MSE}$ part. MSE_0 is the mean squared error (MSE) of the tip data in relation to the phylogenetic mean of the data. MSE is the mean squared error of the tip data in relation to the phylogenetic variance covariance matrix. We explain both of these in more detail below. The important points to take away are:

- if trait similarities are well explained by the phylogeny, observed MSE will be *small* so the ratio of observed MSE_0 to MSE will be *large*, and so will K.
- if trait similarities are *not* well explained by the phylogeny, observed MSE will be *large* so the ratio of observed MSE_0 to MSE will be *small*, and so will K.
- the $observed \frac{MSE_0}{MSE}$ is divided by the expected value under Brownian motion to standardise it because $observed \frac{MSE_0}{MSE}$ will vary depending on the trait and the tree.

K varies continuously from zero to infinity. When K is 0, this indicates that there is no phylogenetic signal in the trait (i.e. close relatives are not more similar on average than distant relatives). This is the equivalent of $\lambda = 0$. When K is 1, this indicates that the trait has evolved according to the Brownian motion model of evolution, just like when λ is 1. Finally, where where K is greater than 1 this indicates that close relatives are more similar than expected under a Brownian motion model of trait evolution (see interpretation notes below). The maximum value of K will vary depending on the trait and the tree involved.

We can test whether K is significantly different from 0 using randomisation tests. These work by randomizing the trait data across the phylogeny, calculating K, and then repeating this 1000 times. If we have phylogenetic signal in our data we expect these random K values to have lower values than our value of K. We can get a p value for the randomisations by counting how many times the randomised trait data gives a higher value of K than our observed value, and then dividing this by the total number of randomisations. For example, if 10 randomised K values are higher than the observed K then p = 10/100 = 0.01.

How do we estimate Blomberg's *K*?

Let's use a very simple example with just three taxa, A, B and C. For more details and proper maths (!) see [@blomberg2003testing].

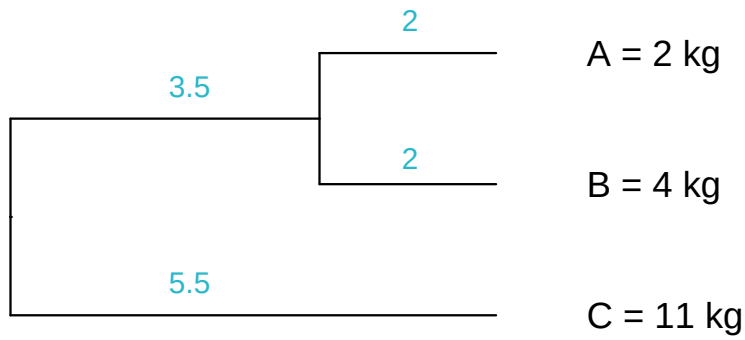

Figure 3.11: Phylogeny showing branch lengths and body size for each tip taxon

As we noted above the equation for Blomberg's K is:

$$K = \frac{observed \frac{MSE_0}{MSE}}{expected \frac{MSE_0}{MSE}} \tag{3.2}$$

Before we think about the equation we should remind ourselves what we mean by *mean squared error (MSE)*. MSE is a way of looking at how far away the data are from the mean. If we have five species with body sizes of 5, 10, 15, 20, 25, the mean body size will be 15 ($\frac{5+10+15+20+25}{5}$). The *error* (E) is how far from the mean the observed values are (this is equivalent to the residuals in a model). Here the errors will be -10 (i.e. $5 - 15$), -5, 0, 5 and 10. If we sum all of these we end up with 0 as the negative and positive numbers cancel each other out, so to solve this problem we *square* (S) them to give 100, 25, 0, 25 and 100. Finally we take the *mean* (M) of this number to get 50. That's the MSE! If the values are clustered around the mean the MSE will be low. If they are spread out the MSE will be high.

This calculation might seem familiar, and that's because it's the same way we estimate variance!

Note that above when we calculated the mean part of MSE/variance we divided by n, where n is the number of observations. However, this is only valid where we are looking at the whole population. If instead, as is common in phylogenetic comparative methods, we are looking at a *sample* of the whole population, we instead divide by $n - 1$. Why do we do this? In simple terms, if we are looking at a sample rather than the whole population, the MSE/variance we calculate will contain a little bit of bias. Subtracting 1 from the sample size corrects this bias, so you'll usually get

a more accurate answer if you use $n-1$ instead of n (if you want more details this is known as Bessel's correction).

Now let's consider just the observed $\frac{MSE_0}{MSE}$ part of the equation for K.

MSE_0 is the mean squared error (MSE) of the tip data in relation to the phylogenetic mean of the data. This is exactly the same as the MSE described above, but instead of using the mean calculated in the normal way, we use the phylogenetic mean, i.e. the mean value if we account for shared evolutionary history of the species involved. A simplified equation for this is as follows, where X is the trait data, a is the phylogenetic mean, and n is the sample size.

$$MSE_0 = \frac{\Sigma(X-a)^2}{n-1} \tag{3.3}$$

The phylogenetic mean of the data is equivalent to the estimated trait value at the root, so we can obtain this using ancestral state estimation. See *Chapter 4* and the online materials for more information on how this works.

For our example, the phylogenetic mean a is 6.6 kg. Values of X are 2, 4 and 11 kg, and n is 3. This means that our MSE_0 is 23.64:

$$MSE_0 = \frac{(2-6.6)^2 + (4-6.6)^2 + (11-6.6)^2}{3-1} = 23.64 \tag{3.4}$$

MSE_0 will vary in magnitude depending on how far the data are from the phylogenetic mean. But in a situation where close relatives tend to be more similar than distant relatives, i.e. the phylogeny gives us some information about the covariances among species trait values, it will be larger than MSE.

MSE is the mean squared error extracted from a phylogenetic generalized least-squares (PGLS) model that uses the phylogenetic variance–covariance matrix in its error structure. We cover PGLS models in *Chapter 4*, but for now it's enough to know that rather than using X we use U which is the trait values transformed according to the phylogenetic variance–covariance matrix.

$$MSE = \frac{\Sigma(U-a)^2}{n-1} \tag{3.5}$$

In a PGLS context, $(U-a)$ are equivalent to the phylogenetic residuals from a PGLS model.

Here the values of U are 7.219543, 4.902944 and 8.633757. a is still 6.6, and n is still 3, thus

$$MSE = \frac{(7.219543 - 6.6)^2 + (4.902944 - 6.6)^2 + (8.633757 - 6.6)^2}{3 - 1} = 3.7 \tag{3.6}$$

As predicted, MSE is much smaller than MSE_0 because it includes the information about the phylogeny.

We can now calculate observed $\frac{MSE_0}{MSE} = \frac{23.64}{3.7} = 6.389189$

Finally, to make values of K comparable among different phylogenies, the observed MSE_0 to MSE ratio is standardized by dividing it by the expected mean squared error ratio under Brownian motion. It is not possible to separately extract expected MSE_0 and expected MSE, but we can get the overall value using the following equation:

$$expected\frac{MSE_0}{MSE} = \frac{1}{n-1} * (\Sigma V_{diag} - \frac{n}{\Sigma V^{-1}}) \tag{3.7}$$

V is the phylogenetic variance covariance matrix, and V^{-1} is the inverse of the phylogenetic variance covariance matrix. For our simple example these look like this:

	A	B	C
A	5.5	3.5	0.0
B	3.5	5.5	0.0
C	0.0	0.0	5.5

	A	B	C
A	0.3055556	-0.1944444	0.0000000
B	-0.1944444	0.3055556	0.0000000
C	0.0000000	0.0000000	0.1818182

In the equation, ΣV_{diag} is the sum of diagonal elements of V:

$$\Sigma V_{diag} = 5.5 + 5.5 + 5.5 = 16.5 \tag{3.8}$$

ΣV^{-1} is the sum of all elements of the inverse matrix of V:

$$\Sigma V^{-1} = 0.3055556 + -0.1944444 + -0.1944444 + 0.3055556 + 0.1818182 = 0.4040406 \tag{3.9}$$

Therefore…

$$expected\frac{MSE_0}{MSE} = \frac{1}{3-1} * (16.5 - \frac{3}{0.4040406}) = 4.537502 \tag{3.10}$$

So overall our K value for this simple example is

$$K = \frac{observed\frac{MSE_0}{MSE}}{expected\frac{MSE_0}{MSE}} = \frac{6.389189}{4.537502} = 1.408085 \tag{3.11}$$

3.4.4 Phylogenetic signal for categorical traits

Both λ and K are only appropriate for continuous traits, but many traits are not continuous, they are categorical. For example, habitat, growth form, colour etc. As an example let's think about frog habitats. Frogs (Anura) can live in a variety of habitats. Some live in trees (Scansorial), some live underground (Fossorial), some live in water (Aquatic), some live on dry land (Terrestrial), and some live in a mixture of dry and wet habitats (Semi-aquatic)

Remember with phylogenetic signal we are looking for the pattern where close relatives are more similar to one another than more distant relatives. For these categories, it would be sensible to say we had high phylogenetic signal if, for example, all toads (Bufonidae) are Semi-aquatic, and all tree frogs (Hylidae) are Scansorial. But what if some toads are Semi-aquatic but others are Terrestrial? If we don't know how species transition from state to state, it's hard to know what we might expect to see as evolutionary distance between species increases. We might sensibly assume that species easily evolve from Semi-aquatic to Aquatic, or from Subfossorial to Fossorial, but what about changes from Aquatic to Fossorial? All of this makes phylogenetic signal for categorical variables tricky.

There are a couple of, more or less satisfying, solutions…

1. Do we actually need to know the phylogenetic signal for these variables? For our models we are interested in phylogenetic signal in the *residuals* (see *Chapter 4* and below), so maybe we don't care about phylogenetic signal in the variables? Unless there is a real need, don't bother! You could visualise what is going on instead by adding colours to the tips of the phylogeny to represent the different categories. This should give you an idea about whether categories cluster in different clades or not.

2. We could code our categories numerically then use λ or K. This is only suitable if the categories are ordered, and if the difference between each pair of categories can be considered equal. For example, a variable that

has low, medium and high values could be coded as low = 1, medium = 2, and high = 3. Again this is not ideal, but will give you an answer. We recommend avoiding this if possible.

3. We could recode these as binary variables and use D (Fritz and Purvis, 2010). For example, creating a new variable called Aquatic, and coding each species as 0 = not aquatic; 1 = aquatic. This is probably the best solution, so we explore D briefly below.

3.4.5 Calculating D for binary traits

The D statistic (Fritz and Purvis, 2010) is calculated as follows:

$$D = \frac{d_{obs} mean(d_{BM})}{mean(d_{rand}) - mean(d_{BM})} \tag{3.12}$$

d_{obs} is the number of character state changes d (from 0 to 1 or 1 to 0) required to obtain the observed distribution of character states at the tips of the phylogeny. The value of d_{obs} will depend on the phylogenetic signal in the variable, but also on the size of the phylogeny and the relative proportion of 0s and 1s. So that we can compare results across datasets and trees, we therefore standardise d_{obs} using two different null models. The first is a *phylogenetic randomness* null model where trait values are randomly assigned to the tips of the phylogeny and then d is calculated. This is repeated 1000 times to produce d_{rand}, the expected distribution of d values if character states are randomly distributed among species without respect to phylogeny. The second null model is a *Brownian threshold* model where we simulate a continuous trait along the phylogeny then define the character state at each tip according to some threshold value of the continuous trait. The threshold is chosen to ensure that the number of tips with each character state remains the same as in the observed data. d is then calculated and the process is repeated 1000 times to get d_{BM}, the expected distribution of d values if character states are distributed among species under the expectations of Brownian motion model of evolution.

Where D is 1 this indicates that the distribution of the binary trait is random with respect to phylogeny, i.e. there is no phylogenetic signal. This is equivalent to λ or K of 0 in continuous traits. Conversely, where D is 0 this indicates that the binary trait is distributed as expected under the Brownian motion model of evolution, i.e. equivalent to λ or K of 1. D can also be less than 0 and greater than 1; where D is less than 0 it indicates that close relatives are more similar than the Brownian motion expectation (equivalent to $K > 1$) and the trait is clustered across the tree, and where D is greater than 1 it indicates close relatives are even

more different than the random expectation, and the trait is overdispersed across the tree.

We can also use d_{rand} and d_{BM} to determine the significance of d_{obs} values. If d_{obs} is *higher* than over 950 of the 1000 d values in d_{rand} ($p < 0.05$), then the trait is significantly more overdispersed across the tree than expected at random. Alternatively, if d_{obs} is *lower* than over 950 of the 1000 d values in d_{BM} ($p < 0.05$), then the trait is significantly more clustered across the tree than expected under Brownian motion.

3.4.6 How do we interpret phylogenetic signal?

Although people will often refer to 'high' or 'low' phylogenetic signal, what people mean by 'high' and 'low' varies widely. For example, in different studies λ of 0.5 might be called low, intermediate or high! This can make interpreting phylogenetic signal problematic. Additionally, as we noted at the start of this section, phylogenetic signal is a *pattern*. Many people however, use it as a way to provide information about evolutionary *process*. Interpreting underlying process from pattern should be done with care, and is often inappropriate.

Phylogenetic signal is a *pattern*, not a *process*. Always be very careful when interpreting underlying process from a pattern. Different evolutionary processes can produce similar values for phylogenetic signal metrics. Phylogenetic signal is also context dependent, and may be influenced by sample size, and error in the tree or trait (see details below).

Traditionally, 'low' phylogenetic signal has been interpreted as evolutionary lability or plasticity (Blomberg et al., 2003), or as the result of high rates of trait evolution leading to large differences among close relatives. Adaptive radiations are expected to be characterized by low phylogenetic signal in ecological niche traits because in adaptive radiations close relatives rapidly diversify to fill new niches. Similarly groups with lots of convergent evolution in a trait are expected to have low phylogenetic signal.

'High' phylogenetic signal is interpreted as genetic drift or neutral evolution, because these processes should approximate a Brownian motion model of evolution (see section above). 'Very high' phylogenetic signal is often interpreted as evolutionary or phylogenetic conservatism (Losos, 2008), where close relatives have extremely similar trait values. The point at which phylogenetic signal is considered high enough to be phylogenetic conservatism varies among authors (e.g. λ or $K > 1$, (Losos, 2008); λ or $K = 1$, (Cooper et al., 2010)). Explanations for phylogenetic conservatism also vary, for example it may occur because species survival requires specific trait values (e.g. traits related to drought tolerance

in desert plants are likely to be highly conserved), or because of other mechanisms such as stabilizing selection, pleiotropy, high levels of gene flow, limited genetic variation, low rates of evolution, physiological constraints or various biotic interactions (e.g. competition) restricting the evolution of new phenotypes (Blomberg and Garland Jr, 2002, Losos (2008)).

Unfortunately, as you may already have gathered, interpreting phylogenetic signal is not as simple as low values equals trait lability and high values equals trait conservatism. Different evolutionary processes can produce similar values for phylogenetic signal metrics, for example, λ or K of 1 may be the result of neutral genetic drift, but the same pattern of traits at the tips of the phylogeny can also be achieved by (1) a model in which traits are evolving to an optimum which itself evolves according to a Brownian process (i.e. the Ornstein–Uhlenbeck model of evolution; see *Chapter 6*); (2) a model where traits are evolving under natural selection that randomly fluctuates; or (3) a model where selective pressures themselves exhibit strong phylogenetic signal, thus producing strong phylogenetic signal in the trait(s) on which these selection pressures are acting (Revell et al., 2008, hansen2008comparative). There is also no relationship between measures of phylogenetic signal and evolutionary rate under simple evolutionary models (Revell et al., 2008).

Phylogenetic signal is also context (i.e. trait data and phylogeny) dependent. The phylogenetic signal in a trait will not be the same at all phylogenetic/taxonomic scales. For example, a trait may have high phylogenetic signal at the family level but this pattern may break down at higher or lower taxonomic levels. Phylogenetic signal can also be influenced by convergent evolution, taxonomic inflation and cryptic species (Losos, 2008). Because of this, using λ or K (or any other measure of phylogenetic signal) to infer evolutionary processes or rates must be performed with consideration of the traits involved, the hypotheses to be tested and any available external information (Cooper et al., 2010, Kamilar and Cooper (2013)).

3.4.7 Some additional caveats with estimating and interpreting phylogenetic signal

We will come back to many of these issues in later chapters, but it's useful to explore these now in the context of phylogenetic signal. When reading this, be aware that these issues will influence all phylogenetic comparative methods in some way. In fact many of these issues are important for any statistical method you use!

3.4.7.1 Sample size.

Small sample sizes tend to have lower *power* to detect significant results. What's power in this context? Statistical power is the probability of a hypothesis test

finding an effect if there is an effect to be found. Large sample sizes, on the other hand, are more likely to yield statistically significant p-values. Consequently, it is important to think about statistical significance versus biological significance when sample sizes are extremely high or low. A non-significant result at low sample sizes may not mean the effect is unimportant, conversely a significant result at high sample sizes does not mean the effect is important. This is really important to think about whatever analysis you're doing.

In the context of phylogenetic signal, K is able to detect significant phylogenetic signal when sample sizes are greater than 20 (Blomberg et al., 2003), i.e. it has good power when there are more than 20 tips in the tree. For λ, power is only good for sample sizes greater than 30 (Freckleton et al., 2002). As the number of tips in the tree increases, the ability to detect significant levels of phylogenetic signal also increases.

For large sample sizes, it may be more useful to focus on the value of λ or K, rather than placing too much emphasis on the significance or non-significance of p-values. For small sample sizes, finding λ or K that is not significantly different from 0 should not be taken to mean that there is no phylogenetic signal in the variable without close examination of the data (Kamilar and Cooper, 2013).

3.4.7.2 Error in the phylogeny and the trait

As we explored in *Chapter 2*, phylogenies are just hypotheses about how species are related, not the truth. Thus all phylogenies contain error. Luckily it seems that missing branch length information has negligible effects on estimates for λ or K (Münkemüller et al., 2012), however, phylogenetic signal may be inflated when using K if the tree has polytomies (this is not a problem for λ) (Molina-Venegas and Rodríguez, 2017).

Trait measurements are also likely to include errors, especially when we use species averages taken from a number of sources (see *Chapter 2*). Simulations have shown that measurement error substantially decreases the power to detect significant phylogenetic signal using K, and also decreases values of K (Hardy and Pavoine, 2012, Blomberg et al. (2003)).

People will sometimes say that they accounted or "corrected" for phylogeny (see *Chapter 1* and *Chapter 4*) because of the phylogenetic signal in their variables. **This is incorrect**. We account for phylogenetic non-independence because the **residuals** from our models show phylogenetic signal (see [@revell2010phylogenetic]). Note that the λ shown in these *phylogenetic generalised least squares* (PGLS) models (see *Chapter 4*) is the λ for the model residuals **not** the individual variables.

Correcting for phylogeny. It is common to come across the term "correcting for phylogeny". However, some feel that by saying that we are "correcting" for phylogeny, we are suggesting that phylogeny is a bad thing that needs to be fixed. In reality, phylogenetic comparative methods take phylogenetic relationships into account, and allow us to incorporate information about the way evolution happens into our models, thus making them more realistic and more useful. Thus we aren't correcting anything. Depending on the company you're in, avoiding the phrase "correcting for phylogeny" is either seen as pedantic, or extremely important. We tend to avoid it, but recognise that for some people thinking of phylogenetic non-independence as error to be corrected in statistical models is very natural and aids their understanding.

[CHAPTER ENDING MATERIAL STYLE STILL TBC BUT WILL BE CONSISTENT ACROSS THE BOOK] ## Summary

3.5 Online stuff

3.6 MCQs

Chapter 4

Dealing with phylogenetic non-independence: a practical introduction.

** RPF is writing this at present **

This chapter will cover the most common use of phylogenetic comparative methods – dealing with phylogenetic non-independence. The focus here is on the key methods that will be used in practice and encountered in the literature. Practicality will be an overarching theme, with worked examples and links to online resources provided to help with this. We will also try to set the methods within wider, more familiar statistical frameworks (e.g. mixed models, other forms of non-independence) so that users can take advantage of pre-existing statistical understanding. Note that this is designed to aid readers familiar with these techniques; if readers are unfamiliar with them it will not substantially reduce their understanding.

- What is phylogenetic non-independence?
- Phylogenetically independent contrasts (PICs) - the historical context of PCMs.
- Phylogenetic generalised least squares (PGLS).
- Related methods e.g. mixed models.
- Links to other kinds of non-independence, e.g. spatial, temporal.

- Emphasising the relationship to familiar statistics, e.g. regression, GLM, mixed models, meta-analysis.

Chapter 5

Testing and evaluating comparative models: is my model good enough?

RPF to write

Like all statistical methods, comparative methods have their limitations. The focus in this chapter will be on succinctly explaining the issue in each case, then using practical examples to show how to deal with the problem. This chapter encompasses a set of issues that commonly arise in comparative studies and have not been reviewed before in the comparative literature.

- Sample size and power analysis.
- Diagnostics for models.
- Model adequacy.
- When shouldn't you use PCMs.
- Critical evaluation of other PCMs (i.e. phylogenetic eigenvector regression).

Chapter 6

More advanced topics, where next?

RPF to write ** May be combined with the previous chapter if appropriate**

This chapter will introduce how we can harness a range of existing powerful statistical tools to add to our comparative methods toolbox. For example, phylogenetic uncertainty can be accounted for using the same statistics used for meta-analysis; missing data theory is well developed in the statistics literature and many tools exist to help deal with this.

- Phylogenetic uncertainty (should you trust the phylogeny): how to quantify?
- Intraspecific variation: when to average, when to not.
- Missing data: does this matter?
- Making predictions.

Chapter 7

Tempo and mode of evolution from comparative data.

NC currently writing

The emphasis in the literature is often on using comparative methods to "fix" statistical tests so that p-values are valid (see Chapter 4). For the past 15 years, however, the cutting-edge of methods development has been on using comparative methods to infer the nature of the evolutionary process: evolution is the reason that we see phylogenetic signal in comparative data, so it makes sense that we should be able to use phylogenies and comparative data to infer the evolutionary process. Here we distinguish the mode (i.e. models of evolution, with discrete and continuous data) and tempo (i.e. rates of evolution) of evolution. The issues we will cover include:

- More complex Brownian models.
- Non-Brownian trait evolution.
- The SLOUCH family of models
- Speciational evolution.?
- Clade dependent rates.?
- Trait dependent rates.?
- Species-interactions.?
- Tree transformations - delta, kappa
- OUwie BM with a trend, multiple rates, OU, OU with many peaks, (read our OU paper again), EB/ACDC

7.1 Brownian motion recap

Remember Brownian motion? If you haven't already read that section of the Foundations chapter, now would be a good time to do so. But if you already feel fairly comfortable, let's just remind ourselves of the summary we produced in that chapter.

The key features of the Brownian motion model are:

- Traits evolve at random, i.e. by a random walk.
- The direction and magnitude of the trait change at one time interval is independent of the direction and magnitude of the trait changes at other time intervals.
- There is no overall drift in the direction of evolution.
- Traits evolve at a rate σ^2.
- The rate of evolution is constant.
- Variance in traits increase (linearly) in proportion to time.
- The overall variance of the Brownian process is the rate of evolution times the amount of time that has elapsed.

In terms of the model parameters:

- The model can be fully described using just two parameters $z(0)$ and σ^2.
- $z(0)$ is the starting value of the trait we are interested in. This is the mean value for the trait seen in the ancestral population before there have been any changes to the trait.
- σ^2 is the rate at which evolution (i.e. trait change) occurs. It is also referred to as the Brownian rate parameter.

Some biological implications of the model are:

- Correlation among trait values is proportional to the extent of shared ancestry for pairs of species, i.e. close relatives are more similar than more distant relatives.
- Different lineages evolve at the same rate.

Hopefully this recap has also reminded you of some of the constraints of Brownian motion as a model of trait evolution…

Chapter 8

Looking forward, more advanced comparative methods and where next.

To write at the end

Here we will briefly describe additional methods we do not cover in the Primer and give interested readers places to go for more information, e.g. diversification analyses.

- A list of more advanced topics, e.g. diversification analyses.
- An overview of current literature and where to look next

Chapter 9

Glossary

To write at the end

- **phylogeny**
- **Root**
- **Branch**
- **Tip**
- **Node**
- **Outgroup**
- **ingroup**
- **Speciation**
- **Extinction**
- **Diversification**
- **Polytomy**
- **sister species**
- **Clade**
- **Monophyly**
- **Paraphyly**
- **Cladogram**
- **Ultrametric**
- **Total evidence trees**

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